When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

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Abstract

Real-time speech systems answer most queries with a local model, but some queries call for a stronger expert. Selective escalation is well studied for text; speech tightens the constraint: the decision must be made before the model commits to a response, and the signal must survive distribution and modality shift. We ask whether a speech model's own hidden states already carry such a competence signal. Our key finding is that they do, but not where escalation methods usually look: the conventional final-layer read collapses under distribution shift, while a mid-network read stays reliable, transfers from text-calibrated probes to speech input, and can be taken at a single end-of-turn position before the first output token, in about 30 ms. A linear probe on this representation gates a live escalation loop: it raises deployed answer accuracy from 42% to 63% while beating matched-rate random escalation, and, with weights, features, and layer fixed, improves all five external speech-QA benchmarks over always-local operation. These results identify a mid-network representation, rather than the final-layer read, as the practical signal for deciding when a speech model should hand off.

1 Introduction

Real-time spoken interaction leaves only a few hundred milliseconds between turns [Stivers et al., 2009]. Systems that meet this deadline keep a local speech model as the talker, so that the audio context and the voice stay in one session, and call a stronger remote model on turns the local model cannot answer. In a text interface this is a standard routing problem. In a speech interface the decision must also arrive before the talker commits to a response; otherwise the system must delay the expert call or retract an answer already under way. The central question of this paper is whether a duplex-trained speech model exposes its own likely failure early enough to hand the turn to a stronger model.

Part of the answer exists on the text side. Correctness of a text language model can be estimated from verbalized self-assessment, output uncertainty, or hidden states [Kadavath et al., 2022, Azaria and Mitchell, 2023, Burns et al., 2023, Kossen et al., 2024], and text routers use query-side or pre-generation features to choose a model [Ong et al., 2024, Varshney et al., 2026, Lugoloobi et al., 2026]. On the speech side, duplex models learn interaction control as a generation policy: special or synchronization tokens decide when the model should listen or speak [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024, Fu et al., 2024, Yu et al., 2024], and recent systems extend the same

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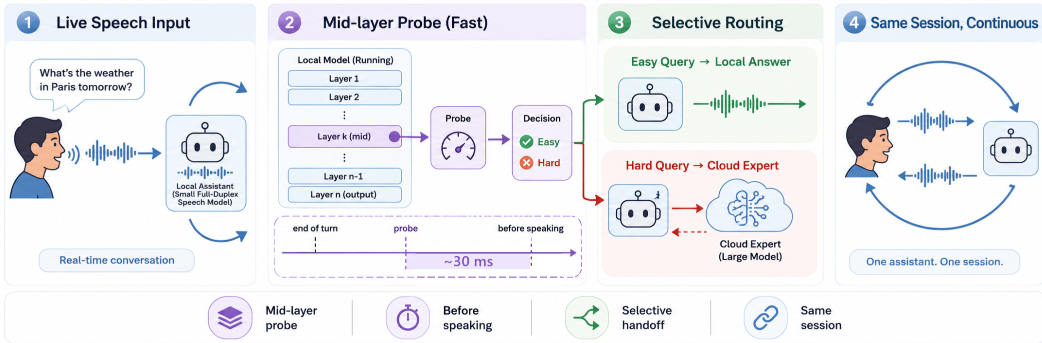

Figure 1: **Primary turn-based system.** **(1)** Speech reaches a duplex-trained local model. **(2)** At the end of the user turn, a mid-network (L22) linear read scores the turn in ∼30 ms, before the first output token. **(3)** Lower-risk turns stay local; higher-risk turns are sent to a cloud expert. **(4)** The local model remains the talker and relays the returned answer. Deployed-channel accuracy rises from 42.2% to 63.3% at 57% escalation in the audio-input/text-output sweep; a TTS-on replay validates the speech path separately.

mechanism to hand work to a reasoning layer or a retrieval service [Huang et al., 2026, Chien et al., 2026].

The trained-control path does not close the gap, for two reasons. First, its cost: the policy is trained into the weights, which takes large amounts of duplex-specific data (hundreds of thousands of hours of synthetic dialogue for SyncLLM, millions of hours of audio for Moshi [Veluri et al., 2024, Défossez et al., 2024]) and risks eroding the backbone's language ability, which is the stated motivation for frozen-backbone designs [Wang et al., 2024b, Yu et al., 2024]. Second, the interface is wrong for escalation: a speak token decides when the model may talk, not whether its answer will be right, and it arrives as a binary action at the moment of commitment. Even when the token triggers a reasoning step instead of speech [Huang et al., 2026], the decision is still a trained binary action: it offers no calibrated score to threshold at a fixed expert budget, and the policy cannot change without retraining. What remains unknown is whether the escalation decision needs training at all: whether a frozen duplex-trained checkpoint already carries a competence signal that can be read before response commitment. Text-probe results make this plausible, but they do not say where such a read stays reliable after duplex speech training, whether a probe calibrated on inexpensive text data still reads speech-input states, or whether the read arrives early enough to act on.

These three conditions are exactly what a deployment needs: a score that separates individual failures rather than memorizing the query types seen in calibration, that survives the text-to-speech shift, and that improves which turns are escalated at a fixed call budget. If such a signal exists, any frozen duplex model gains selective escalation for the cost of a linear probe; if it does not, speech escalation inherits the training costs above.

We test these conditions on MiniCPM-o 4.5 [Cui et al., 2026], a 9B omni model with full-duplex training, against matched controls: raw backbones, streaming-omni and audio-only fine-tunes, a frozen-backbone adapter, and a second duplex generation. The study uses judged failure labels on 600 public-benchmark queries in five frozen pools, leave-one-pool-out (LOPO) transfer as the primary robustness test, matched text and speech inputs, and a complete live escalation loop. We compare decode-time uncertainty, verbalized self-assessment, and linear hidden-state probes across layers, then freeze the winning read for evaluation on public speech benchmarks. Model checkpoints stay frozen; only linear probes are fit on stored activations. Throughout, duplex-trained refers to the weights; the primary live evaluation is turn-based, with audio input and text output, and the concurrent and native full-duplex serving regimes are validated separately (Appendices H and O).

This study makes three contributions:

1. **Transfer moves the readable layer.** The conventional final-layer probe holds in distribution but inverts on held-out math (AUC 0.372); the same last-token read at a mid-network layer reaches 0.931. Matched comparisons tie the late-layer failure to the duplex checkpoints we test; its mechanism remains open (Appendix D).
2. **The mid-layer read satisfies the deployment conditions.** A text-calibrated probe at the same layer reads speech-input states, reproduces on real human recordings, and is available at the end of the user turn in $\sim$30 ms, before the first output token.
3. **The read is sufficient to drive escalation.** In the live loop it raises deployed-channel accuracy from 42.2% to 63.3% at 57% escalation, beats matched-rate random escalation at every tier, and dominates the model's own built-in reasoning mode on both accuracy and latency (Appendix F). With its layer and features frozen, the gate improves all five external speech-QA benchmarks over always-local operation, although selectivity over random is limited on some pools.

The claim is therefore about readout rather than mechanism: the model does not explain why it will fail, but its likely failure is more reliably readable mid-network than at the conventional output interface, early enough to support a live handoff.

2 Related Work

Pre-answer competence estimation. Text-language-model correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], output uncertainty, or hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024]. Intermediate layers can carry more useful information than the output distribution [Orgad et al., 2025], and trained hidden-state probes are strong factual-confidence estimators [Mahaut et al., 2024]. Semantic entropy improves uncertainty estimation by comparing sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]; Semantic Entropy Probes amortize this signal into a pre-generation or post-hoc hidden-state read [Kossen et al., 2024]. Attention-map probes and multimodal uncertainty fusion extend the same broad idea [Chuang et al., 2024, Zhang et al., 2026a]. This work establishes that competence is readable in text models. It does not show where the read remains reliable after duplex speech training or whether a probe calibrated on text transfers to speech-input states.

Selective routing. Text routing splits into two families by where the decision lives. In the first, the decision is read from outside the generator: HybridLLM [Ding et al., 2024], RouteLLM [Ong et al., 2024], and query-side routers [Xue et al., 2025, Hu et al., 2024, Li et al., 2026] choose a model for a complete text query from query features or preference data; FrugalGPT organizes cascades that can score an earlier model's response [Chen et al., 2023], while AutoMix verifies a drafted answer before escalating [Aggarwal et al., 2024]. Closest to our signal, recent methods read pre-generation activations or fit linear success probes on frozen states [Varshney et al., 2026, Lugoloobi and Russell, 2025, Lugoloobi et al., 2026]. In the second family, the decision is trained into the generator: Toolformer and Self-RAG learn tool or reflection tokens [Schick et al., 2023, Asai et al., 2024]; Co-LLM and CITER learn token-level deferral [Shen et al., 2024, Zheng et al., 2025]; and pause, thought, or think/no-think tokens allocate more internal computation [Goyal et al., 2024, Zelikman et al., 2024, Yang et al., 2025, DeepSeek-AI et al., 2025]. Learning-to-defer and early-exit methods make related decisions within a model [Madras et al., 2018, Schuster et al., 2022]. Our contribution is not the linear pre-generation probe by itself. We test whether this kind of read survives duplex training and audio input, and use a deployment-time threshold without modifying the speech checkpoint.

Duplex control and live handoff. End-to-end spoken dialogue includes native duplex models [Nguyen et al., 2023, Défossez et al., 2024, Zeng et al., 2024, Fang et al., 2026], streaming omni models [Xu et al., 2025, Cui et al., 2026], frozen-backbone adapters [Wang et al., 2024b], and audio fine-tunes [Chu et al., 2024]. Most control work in these systems concerns the floor: state, idle, synchronization, or interruption tokens [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024, Ma et al., 2024, Fu et al., 2024, Yu et al., 2024], structured action channels [Zhang et al., 2026b], or classifier heads over a frozen or lightly adapted backbone [Wang et al., 2024b, Chen et al., 2025, Liao et al., 2025]. Training this control is costly enough that reducing the cost is itself a research goal [Xu et al., 2024], and trained turn-taking remains unreliable across models [Lin et al., 2025]. These mechanisms decide when the model may speak; even the most recent native design adds only a few turn tokens to the vocabulary and answers every turn locally, whatever the model's competence on that turn [Fang et al., 2026].

The direct neighbors of our gate are trained think triggers. DuplexOmni emits a think-control token at the end of a user utterance to hand work to a separate reasoning layer [Huang et al., 2026]; chronological thinking interleaves reasoning with the speech stream [Wu et al., 2026]; MiniCPM-o 4.5 ships a built-in reasoning mode [Cui et al., 2026]; MoshiRAG triggers asynchronous retrieval from the model's emerging output [Chien et al., 2026]. These systems show that a live handoff is possible. But the trigger is trained into the weights from purpose-built duplex data, so changing the policy or the downstream resource means retraining; it is a binary action with no calibrated score, so the escalation rate cannot be set to a call budget; and the trained policies are hard to reuse or compare, because released artifacts often include the training recipe but not the trained checkpoint [Huang et al., 2026]. None of these works evaluates a competence score read from a frozen checkpoint before generation. Our probe replaces the trained trigger with such a read: it needs no training of the speech model, exposes a thresholdable score, and on our target model it dominates the built-in reasoning mode on both accuracy and latency (Appendix F).

Positioning. Prior work therefore establishes three ingredients separately: competence can be read from text models, text queries can be routed, and duplex models can coordinate speaking or external computation. To our knowledge, the reviewed literature does not establish their conjunction: whether a frozen duplex-trained speech model exposes a pre-answer competence signal that survives unseen query pools and text-to-speech transfer, and whether that signal arrives early enough to improve live selective escalation. We evaluate these conditions directly. The claims are about readout robustness and system utility; the training mechanism behind the late-layer failure remains open.

3 Reading Competence from Hidden States

This section establishes where and when a competence signal can be read from the frozen checkpoint. The analyses use the target model and matched controls introduced above: 600 public-benchmark queries in five frozen pools with judged failure labels, and logistic probes fit on stored hidden states. Full details of models, data, labels, and probe fitting are in §4.

3.1 Signal comparison under distribution shift

Table 3 compares the candidate signals on MiniCPM-o 4.5. In distribution the final-layer probe looks healthy (AUC 0.822), but most of that accuracy reflects query type rather than instance difficulty: the same hidden state predicts which pool a query came from, a classifier given only the pool identity recovers most of the probe's margin, and appending true pool labels to the probe's features adds nothing. The same decomposition holds at scale on the deployed calibration mix (Appendix O). We therefore treat leave-one-pool-out (LOPO) transfer as the primary metric, and under it the picture reverses: with text input the final-layer read falls to AUC 0.372 on held-out math, and when the trap pool is held out the probe scores it on the confident side even though the model fails every query in it. A router built on query text shows the same dependence on pool structure and also degrades under LOPO, with little improvement from substantially more data (Appendices B, E). In-distribution routing accuracy alone therefore cannot establish instance-level transfer.

Verbalized self-assessment transfers differently. Asked before answering, $p(\mathrm{True})$ catches exactly the trap pool the probe misses (0.945). Asked after drafting, the model endorses its own confident-wrong

Table 1: Main results: selective escalation on the internal test set and five public speech-QA benchmarks (post-relay answer accuracy, %, audio-input/text-output live loop). The probe is fixed; tier thresholds are label-free per-pool quantiles (the internal tiers realize 14/31/57% escalation). Matched-rate random mixes the always-local and measured always-escalate arms at each pool's realized rate. avg and Δ (average gain over always-local) cover the five external pools. The bottom block applies the same recipe to NemotronLabs-VoiceChat-11B offline, re-mixing cached local and expert outcomes (Appendix C). AlpacaEval, reported separately on its 1–5 scale, changes from 3.99 to 4.35 on MiniCPM but does not beat matched-rate random; the second family changes from 3.55 to 4.46 versus 4.23 for random. Per-tier random references and gold-input ceilings: Appendices I–J.

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	Reason. zh	avg	Δ
live post-relay view (audio input, text output)								
always-local	42.2	66.4	56.8	83.6	51.5	58.9	63.4	—
+ gate @15%	52.8	73.2	62.8	90.4	60.0	65.3	70.3	+6.9
+ gate @30%	59.2	80.0	68.0	90.4	72.0	71.3	76.3	+12.9
+ gate @50%	**63.3**	**86.0**	**73.2**	**94.8**	**79.5**	**77.2**	**82.1**	**+18.7**
matched random @50%	56.1	77.2	68.8	88.6	70.0	71.3	75.2	+11.8
always-escalate (measured)	66.5	88.0	80.8	93.6	88.5	83.7	86.9	+23.5
always-expert (gold-text ceiling)	92.2	96.8	86.4	92.8	93.0	87.1	91.2	+27.8
second family: NemotronLabs-VoiceChat-11B (offline re-mix, official judges)								
always-local	—	35.6	37.6	70.5	31.0	—	43.7	—
+ gate @15%	—	49.8	48.0	78.6	42.5	—	54.7	+11.0
+ gate @30%	—	61.5	58.8	83.8	54.5	—	64.6	+21.0
+ gate @50%	—	**74.1**	**69.6**	**87.6**	**67.5**	—	**74.7**	**+31.0**
matched random @50%	—	65.6	60.2	81.8	60.0	—	66.9	+23.2
always-expert (ceiling)	—	95.5	82.8	93.2	89.0	—	90.1	+46.4

answers and the signal inverts: post-draft $p(\mathrm{True})$ is the strongest single readout in distribution but no longer provides a pre-answer routing signal.

3.2 Layer and position sweep

Matched comparisons tie the failure to the duplex checkpoints rather than to the backbone alone. The raw Qwen backbone, subsampled to match MiniCPM-o 4.5's per-pool failure rates, holds LOPO math near 0.96 where the duplex checkpoint scores 0.372, and the Qwen2.5 family shows the same ordering, raw above streaming-omni above duplex (Table 4, appendix). The Qwen2-Audio comparison is not attribution evidence because the audio fine-tune also changes task capability and the failure distribution substantially (Appendix B). A layer $\times$ position sweep localizes the difference (Figure 2): it is concentrated at the last-token position in late layers on text input, mean-pooled reads degrade far less, and at L22 the last-token probe reaches **0.931** on the same held-out math pool, close to the raw backbone.

The failure is input-specific rather than universal: with audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. We use L22 because one read point must remain robust for shifted text queries and support text-to-speech transfer, not because the final-layer signal is absent on the model's native audio input.

The sweep establishes where the transferable read survives, not why it weakens late. Direct tests do not support the simple hypothesis that turn-control state takes over the last-token representation: the layer at which performance drops does not move in speak mode, and the inverting probe direction aligns with neither the control vocabulary nor an explicit listen/speak direction. A decomposition further shows that a distributed transferable component weakens over the final layers of the two duplex checkpoints but not their raw backbones. These observations constrain the explanation but do not identify a training objective or causal mechanism; the full audit is in Appendix D.

The mid-network pattern also appears outside the MiniCPM family: on NemotronLabs-VoiceChat-11B, a hybrid-Mamba duplex model, the same recipe peaks mid-network and transfers to the external pools from a 600-query calibration set (Appendix C).

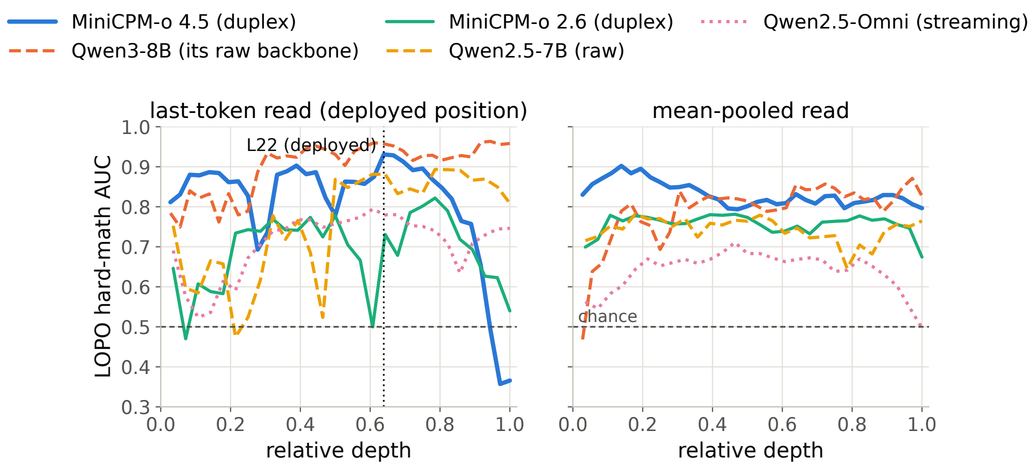

Figure 2: The core representational result: LOPO math AUC by layer and position. On the duplex model the standard (final-layer, last-token) read inverts on text input while mid-network layers hold $\geq$0.9; raw backbones show no cliff. The deployed gate reads L22.

3.3 Modality transfer

Probes trained on text hidden states and scored on speech-input hidden states show where the representation becomes shared across modalities: early layers are modality-specific, and from about half depth the transfer plateaus (0.855 at the deployed L22). This replicates on MiniCPM-o 2.6 and cross-family on Qwen2.5-Omni. The decisive control is Freeze-Omni: identical frozen LLM weights with audio attached by an adapter, and cross-modal transfer stays far below the end-to-end models. Text calibration therefore transfers in the end-to-end speech models tested here, but not in the adapter-attached control. Verbalized introspection shows the complementary failure: on audio input, pre-answer $p(\text{True})$ collapses on both MiniCPM duplex generations while both non-duplex controls stay intact. Controlled arms show the verbal self-check depends on the text-token pathway (duplicating the question as text restores it), while a linear probe remains usable on the same forward pass, and the restored self-check is additive with the probe (Appendix B, Table 7). The mid-layer band also replicates on 200 real human recordings from SD-QA, so the result is not limited to the TTS rendering. The live system keeps L22 and adds two pooled positions to form the deployed probe (Appendix B).

3.4 Decision timing and escalation budget

Scoring latency. The gate is a linear read on L22 states the forward pass already computes: a 4096-d dot product per position (the deployed probe adds two pooled positions at no added scoring latency; Appendix B). On an H100, measured from prefill start with CUDA synchronization, the score is available at P50 in 20 ms for text and 45 ms for audio, against a time to first token of 36/68 ms on the same benchmark. The measurement excludes network time, but it establishes the ordering the system needs: the cloud request can start before local decoding commits to an answer. Waiting for decoded tokens does not improve the read in our data and postpones the expert request (Appendix F).

Read position in the audio loop. In the streaming loop, scores taken from intermediate audio chunks are weaker than a read at the end of the turn and can miss information in the last chunk. The system therefore reads once, at the end of the user turn: after the final audio chunk it prefills the assistant turn and reads L22 at its start. In a 360-query replay, this single read reaches AUC 0.887 in about 30 ms, against 0.843 for the offline reference, and arrives before the first output token (Appendix G).

From score to escalation budget. Thresholding the score converts it into a call budget. On the frozen test split, the probe beats matched-rate random escalation at every operating point and recovers over half of the local-to-expert gap with a third of the expert calls (Figure 9, appendix). In distribution it is statistically tied with a pool-identity classifier, as §3.1 predicts; its advantage appears under transfer, where pool identity stops being sufficient. The gate is fit to local failure, not expert benefit: a benefit label would specialize the ranking to the current expert. The frozen scores remain useful under the stricter benefit label, but pools with little expert headroom remain a system-level limit (Appendix I). Comparisons with the model's built-in reasoning mode are in Appendix F.

4 Experimental Setup

4.1 Models

The target model is MiniCPM-o 4.5 (9B; omni + full-duplex fine-tune of Qwen3-8B), run in bf16 with greedy decoding and seed 42. To attribute effects to the kind of fine-tune rather than the backbone, we assemble ten models spanning raw backbones, a second duplex generation (MiniCPM-o 2.6), a streaming-omni fine-tune (Qwen2.5-Omni-7B), an audio-only fine-tune (Qwen2-Audio-7B), and a frozen-backbone adapter model (Freeze-Omni, LLM weights byte-identical to its raw control); the full roster is Table 2 (appendix). The comparison statistic is always the within-pair difference. All model checkpoints remain frozen throughout; the escalation expert is `gpt-5.5` at low reasoning effort.

4.2 Dataset and labels

The internal dataset is 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test): *trap* = SimpleQA [Wei et al., 2024] (the target fails 100%), *hard-knowledge* = MMLU-Pro [Wang et al., 2024c], *easy-fact* = TriviaQA [Joshi et al., 2017], *easy-chat* = Dolly/Alpaca [Conover et al., 2023, Taori et al., 2023], *hard-math* = GSM8K tail + MATH-500 [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]; public benchmarks only, no LLM-generated queries. Adequacy labels come from an LLM judge [Zheng et al., 2023] with structured outputs, blind to the gate and validated by a stronger re-judge (agreement 96.2% text / 94.5% audio; every headline number moves $\leq$1 point under the stronger labels). The audio arm renders the same frozen pool to speech with single-voice TTS (content unchanged; only modality varies), following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]. Judge prompts, pins, and spend: Appendix M–N.

4.3 Probe training and baseline signals

We compare three signal families: (i) decode scalars: max token entropy or margin over the first k decode steps; (ii) linear probes: logistic regression on hidden states, with layer and position selected on calibration data; and (iii) verbalized self-evaluation [Kadavath et al., 2022]: $p(\text{True})$-pre asks "would you answer this correctly?" before answering, $p(\text{True})$-post asks about a drafted answer; $P(\text{Yes})$ is read from the first token's distribution. The representation analyses use a single-position, 4096-d L22 probe fit on the 360-query calibration split. The live system uses the same layer plus two pooled positions, fit on a wider 2,310-query calibration mix that is disjoint from the external benchmarks. Concurrent and native serving change the read point, so those validations refit the same feature set in regime; Appendix B records the full calibration ledger.

4.4 Benchmarks and evaluation protocol

Audio-side findings are validated on real human recordings (SD-QA, Faisal et al., 2021); the frozen gate is evaluated on external public speech benchmarks with their official audio (§5.2). Following §3.1, we use LOPO AUC to test distribution transfer and text-trained/audio-scored AUC to test modality transfer. For the live system, the key comparison is against matched-rate random escalation: an improvement over always-local operation can come from the expert alone, while an improvement over random at the same call rate measures the value of the gate's ranking.

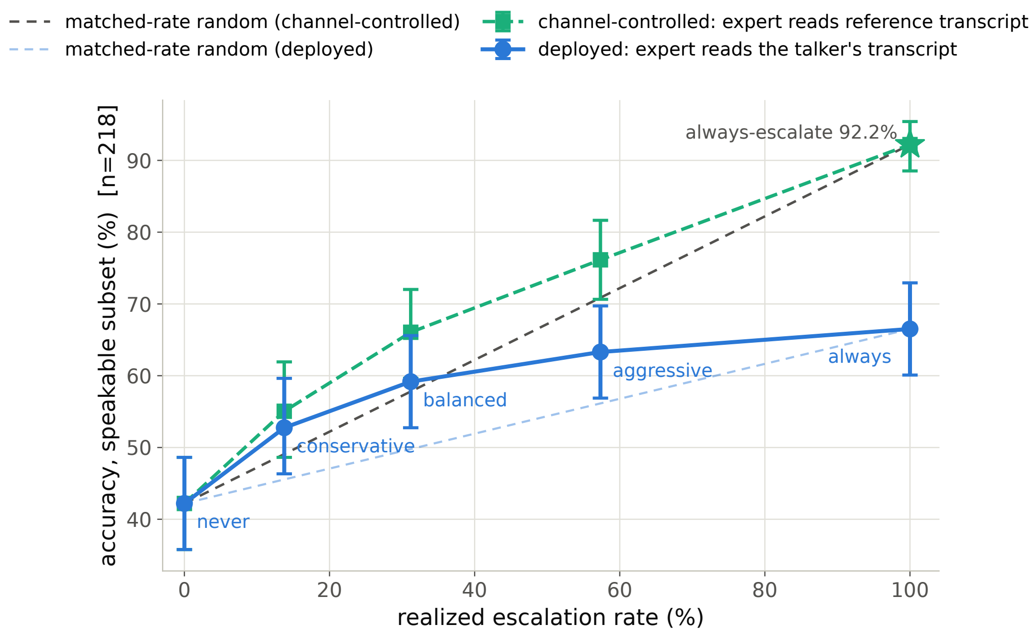

Figure 3: Live escalation on the internal test set (speakable subset, n=218). Blue: deployed-channel accuracy, where the expert answers from the talker's own transcript. Green: the same sessions re-scored with the expert answering the reference transcript, which factors out the relay channel. Each curve has its own matched-rate random reference (dashed, same floor and endpoint). Selection beats random in both, by 7.2–9.4 points deployed and 5.3–8.3 points channel-controlled; the gap between the curves is the cost of the relay channel.

4.5 Live loop and escalation path

The live loop is turn-based: audio streams in 1 s chunks, and the talker speaks only after the user's turn ends; the model's native listen/speak head is not engaged, which isolates competence handoff from turn-taking (native-head validations are reported in §5.1 and Appendix O). The loop uses the deployed three-position probe of §4.3 (frozen-test AUC 0.879). Escalation runs in four tiers from conservative to aggressive; each tier's threshold is a quantile of the probe's score, so a tier sets the call budget without correctness labels. If a turn escalates, the talker first produces a short stall phrase; the model's own transcription of the query is sent to `gpt-5.5`, and the returned answer is injected as context for the talker to relay. The expert request runs asynchronously. Partial-query prefetch, expert reasoning effort, and injection controls are in Appendix G.

5 Results

5.1 Live escalation

We ran the 240 frozen test queries through four escalation tiers, each a complete turn-based loop with audio input (§4.5). The primary sweep retains the model's output as text; a separate run of the balanced tier with speech synthesis enabled leaves all 84/240 gate decisions unchanged and measures speech onset. We score deployed-channel answer accuracy: the correctness of the relayed content after the full system path. Headline numbers use a pre-registered input filter that removes 21 formula-LaTeX queries and one broken recording, because the speech rendering destroys the symbolic input before the model receives it. Accuracy rises monotonically, $\mathbf{42.2}\% \to \mathbf{52.8}\% \to \mathbf{59.2}\% \to \mathbf{63.3}\%$ at 0/14/31/57% escalation, and every tier's gain is significant (paired bootstrap; Table 1, Figure 3). The gate also exceeds matched-rate random escalation at every tier: by 7.2–9.4 points on deployed-channel accuracy, and by 5.3–8.3 points when the relay channel is factored out (Figure 3).

A measured always-escalate arm reaches 66.5%. The gain from 57% to 100% escalation is small because the expert sees the model's own transcription: the same sessions reach 92.2% when the expert receives the reference transcript, so the transcription channel, not selection, is the limit at high escalation rates. Live accuracies have a replication floor of $\pm$2–3 points; no claim rests on a smaller difference. Full tables, latency profiles, and boundary cases are in Appendix G.

Serving-regime checks. Replaying the loop with the talker enabled leaves the frozen ranking nearly unchanged (AUC 0.871/0.856 versus 0.866 with the talker disabled). A teacher-forced concurrent-prefill variant is less stable and requires an in-regime refit; its external results remain exploratory (Appendix H). Under the model's native full-duplex head, the read moves to the head's own listen$\rightarrow$speak commit point, and the probe is refit in regime on a widened 5,228-row calibration merge under the official serving configuration, with labels judged on the native answers. The refit reads at AUC 0.876 internally and 0.771 on the four English external QA pools. The Mandarin reasoning pool is lower, at 0.621; a 1,500-question held-out Mandarin set of facts, knowledge, and commonsense reads at 0.811, so the gap is multi-step reasoning rather than language (Appendix O). Re-mixing the native gate's decisions with cached expert outcomes beats matched-rate random on all five English pools at both deployed tiers; with per-language thresholds the Mandarin pool escalates at the nominal rates and beats matched-rate random at the two lower tiers. These checks support the ranking beyond the primary turn-based loop, but they also show that read points, probe weights, and thresholds must be calibrated for the serving regime (Appendix O).

Native full-duplex live sweep. We then re-ran the full five-arm live protocol as native full-duplex sessions on all seven pools (one fresh duplex session per query, official serving configuration, deployed gate, real transcription uplink and expert call, wait paced in wall time). The first pass exposed the relay, not the gate, as the bottleneck: on Speech TriviaQA the expert's returned text scored 0.960 under the official judge, but what the talker voiced after a steering prompt scored 0.728 (never-escalate 0.628), because the head truncated, ignored, or re-answered the relayed text on about a quarter of turns. Speaking the cleaned expert text verbatim through the talker's own teacher-forced voice, with a context note in place of the steering prompt, closes that gap (same 60 queries: 0.733 versus 0.933) at about two seconds of added synthesis time, and is now the deployed relay. With it, native local floors sit 1–8 points below the turn-based loop, but the routing gain is larger: at the aggressive tier delivered accuracy rises from 0.628 to 0.884 (56% escalation) on Speech TriviaQA, 0.528 to 0.752 (60%) on Speech WebQ, 0.480 to 0.825 (66%) on SD-QA, and 0.510 to 0.728 (50%) on the Mandarin reasoning pool, which now escalates at its nominal rates; Llama Questions gains 3 points at 19% escalation, and the turn-based "selective beats always" reversal on that pool does not reproduce natively. Local decoding is stochastic, and a zero-escalation re-run moves a floor by up to 3.6 points, so sub-5-point differences are within noise. Open-ended generation remains the boundary: on AlpacaEval the always-escalate arm scores 4.20 against 4.96 for the expert text, because a 400-character spoken cap removes half of the long answers. Classifying the 600 native failures by cause shows where the pre-answer read works: specific-but-wrong answers are 70% of failures and are ranked at AUC 0.81 (79% escalated at the aggressive tier against a 31% false-fire rate on correct turns), misheard questions at 0.89, whereas execution slips in multi-step reasoning are at chance (0.47) — the error has not happened yet when the gate reads, so only a later read point could recover them.

5.2 Transfer to public speech benchmarks

The shortcut analysis of §3.1 makes external transfer necessary. We freeze the probe weights, feature set, and layer selected on the internal calibration mix, then run the live loop of §4.5 on six public speech benchmarks with audio input and text output: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] (spoken renderings of Joshi et al., 2017, Berant et al., 2013, and the set of Nachmani et al., 2024), SD-QA (real dialectal human speech) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All audio is the benchmarks' official pre-rendered audio; no evaluation query is used to fit the probe. The only per-pool adjustment is label-free: tier thresholds are quantiles of that pool's score distribution, requiring no correctness labels or judge calls. This controls the call budget but cannot improve ranking. A single absolute threshold also beats matched-rate random on four of five QA pools, but its escalation rate varies from 2% to 48% (Appendix I). We use each benchmark's official judge when available and verify that the local-only controls are on the published scale.

The frozen scores remain discriminative outside the calibration mix: AUC ranges from 0.683 to 0.806 across the five QA pools, with a mean of 0.771 (Appendix J). The gate also improves deployed-channel accuracy over always-local operation on every QA benchmark and at every budget; at 50% escalation, mean accuracy rises from 63.4% to 82.1%. These gains establish system utility but do not by themselves establish selectivity, because random escalation also adds expert answers.

Llama Questions provides the cleanest fixed-budget test. It contains a single query type, the kept-local subset scores 97.6% while the escalated subset scores 69.6% (z=6.46), and expert answers improve the latter to 88.8% (p<.0001). At 50% escalation, the measured system reaches 94.8%, compared with 93.6% for always escalating. Matched-rate precision reaches 95% of its base-rate cap (Appendix J). Cross-lingual transfer also improves when the English calibration mix is widened: Chinese Reasoning QA rises from AUC 0.621 to 0.683 despite adding no Chinese calibration data.

The transfer results also expose the boundary between predicting local failure and creating expert benefit. SD-QA is a positive real-speech case and exceeds matched-rate random at all three budgets. Web Questions is less stable, with little gain over random at the low and high budgets; in its top-scoring 15%, many local failures are also missed by the expert (Appendix I). AlpacaEval is a clearer negative: its open-ended failures do not present the retrieval-failure pattern the probe reads. On the second duplex family, which has more expert headroom on these pools, the same recipe beats matched-rate random on all five evaluated pools, including AlpacaEval ($p \leq .0003$). Thus external utility depends on both the quality of the competence ranking and whether the selected failures are repairable by the expert (Appendices C, I, and J).

6 Discussion

Interpretation. The competence signal this system runs on was already in the frozen weights. Nothing was trained into the speech model, and the read costs one dot product on states the forward pass computes anyway, so a deployed duplex model can gain selective escalation without new duplex data, a new control token, or a retrained policy. What matters is where the read is taken. The final layer is the default choice because it is the interface to the output, and on these duplex checkpoints it is the wrong one: the same linear read that inverts at the output holds up in the middle of the network. This also points back at duplex training itself. The matched comparisons show that the fine-tune, not the backbone, is what breaks the late-layer read, so a training recipe that adds interaction control may be degrading the model's own competence interface while leaving its answers intact. We do not know the mechanism, and a model that reads its own state before answering still cannot explain why it will fail.

Scope of the signal. The probe is strongest on retrieval-like failures, where the state reflects a missing answer before generation. It is weak on false premises and complementary to signals that appear during decoding (Appendix G). This boundary explains why failure AUC and system benefit are not the same quantity. A correct ranking cannot help when the expert fails on the same turns, and an open-ended quality benchmark may not contain the factual-retrieval event the probe detects. The unstable Web Questions curve and the negative AlpacaEval result are part of that boundary rather than exceptions to it; SD-QA, by contrast, confirms selective gains on real human speech. Selective escalation needs both a readable local failure and expert headroom.

Calibration. The model weights stay frozen, but the gate is still a fitted component. Text-only calibration is enough for the cross-modal result on end-to-end speech models, the deployed probe uses a wider benchmark-disjoint mix, and both the concurrent and native read points require refitting in their own regime. The method avoids fine-tuning the talker and moves the operating budget through a threshold, but it does not remove the need for representative calibration data.

Limitations. The internal study uses one phrasing per $p(\mathrm{True})$ variant, and its real-speech validation is scripted read speech in a single dialect. Formula-symbolic content is out of scope for the audio arm, since any speech rendering destroys it. Labels come from one judge family, with no human labels; a stronger re-judge bounds the resulting shifts at about one point. Each query is run once per tier, which puts a replication floor of two to three points on every live accuracy, and no claim here rests on a smaller difference. The second-family replication covers mid-band readability and frozen-recipe transfer only, not matched-pair attribution, the live loop, or the concurrent regime. Layer and features were chosen by cross-validation on the internal calibration split and then confirmed on held-out pools, rather than by nested selection inside each held-out pool. The mechanism behind the late-layer decay on text input remains unresolved (Appendix D).

Serving regime is the sharpest practical limit. Calibration does not carry across regimes: under concurrent prefill the frozen read degrades and needs an in-regime refit, and the native read needs its own refit, wider calibration, and labels judged on native answers before it matches the turn-based loop (Appendices H, O). That native read also appears saturated, in that more labels of the same kind, larger heads, and a distilled second-signal student all leave it close to where it is. Finally, the evaluation is scripted and single-turn. Disfluent correction, chained tool calls, and multi-turn grounding remain open [Lin et al., 2026], and in native testing a new question asked during the expert wait usually derails the pending relay, so a production system has to cancel or replace an in-flight request when the user moves on.

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A Model roster

Table 2: Model roster. "Pair" groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

B Extended signal and readout analyses

Table 3: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix B.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	≈chance on raw backbones	—
$p(\text{True})$-pre	0.807	trap 0.945 (probe: 0.232)	**collapses**
$p(\text{True})$-post (full draft)	**0.899**	—	degrades
final-layer probe	0.822	**0.372** (inverts)	0.936 (intact)
mid-layer (L22) probe	0.866	**0.931**	**0.855** (text-calibrated)

The in-distribution shortcut, quantified. The numbers behind §3.1's claim that the final-layer probe's in-distribution AUC of 0.822 is mostly pool identity: a linear read of the same hidden state classifies a query's source pool at 95.8% accuracy; a pool-identity oracle alone reaches AUC 0.715; and appending true pool labels to the probe's features moves it 0.822→0.821. In-distribution area over random is statistically on par between the gate and the pool oracle (Appendix E); the two separate only under LOPO, where the probe inverts to 0.372 and the oracle is undefined.

ROC curves. Figure 4 shows the calibration-split ROC for the signals of Table 3.

$p(\text{True})$ **backbone dependence.** Across backbones, post-draft $p(\text{True})$ beats the trained final-layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone's self-evaluation quality.

Matched pairs. Table 4: with label coverage matched by subsampling the raw control to the duplex model's per-pool fail rates, the raw backbone keeps LOPO math at 0.825/0.809 while duplex models sit at chance: label coverage is ruled out; the representation changed. Degradation is ordered raw $>$ omni-streaming $>$ duplex. Table 5 gives the by-depth summary behind Figure 2: the within-pair deficit at the best mid-layer is only -0.03 vs. -0.59 at the final layer; mean-pooled probes survive to the final layer on duplex models (LOPO math ≈0.80 at L35 on o4.5, 0.674 at L27 on o2.6, 0.13–0.15 below each model's mid-layer last-token peak, and uniformly weaker in-mix, -0.058 at L22); duplex damage is severe but local, omni-streaming damage mild but diffuse.

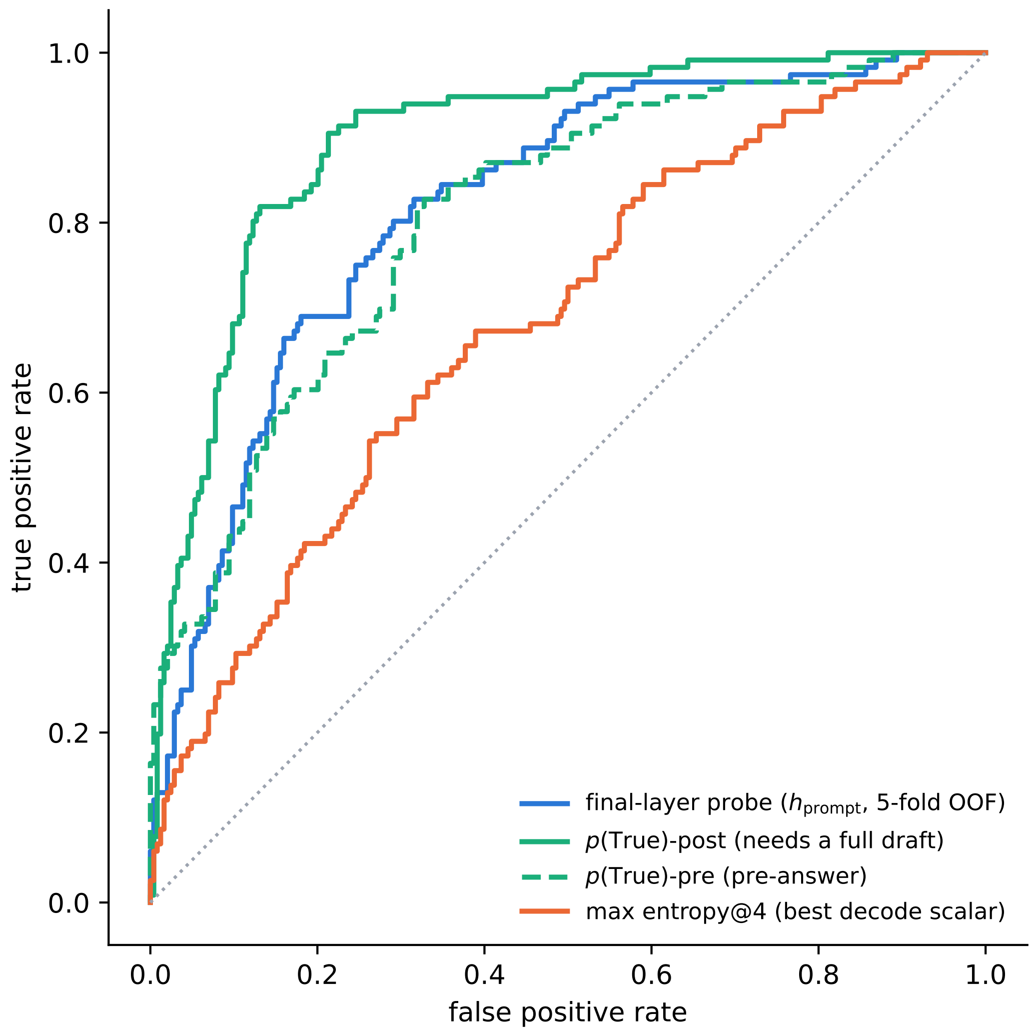


Figure 4: ROC on the calibration split. $p(\text{True})$-post leads but needs a full drafted answer; the probes and $p(\text{True})$-pre are pre-answer.

Table 4: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	**0.809**	0.745	**0.538** $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	**0.526** $\approx$ chance	0.61

Modality specificity of the cliff. Under audio input the final layer does not invert (L35 audio LOPO math 0.936 vs. 0.366 text); a speak-mode template control on text input leaves the cliff unchanged (0.362): the operative variable is the modality of the input context, not the output mode.

Audio collapse of verbalized introspection. Table 6. Three controlled arms pin the mechanism on MiniCPM-o 4.5: not perception (ASR audit WER 0.074; $p(\text{True})$ on the model's own transcript snaps back to 0.074; the well-heard subset still collapses); not a context prior (irrelevant filler audio does not inflate p_{yes}); binding (duplicating the question as text alongside the audio fully restores introspection, 0.034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at 0.93+.

Table 5: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	**0.931 (L22/36)**	**0.366**	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	**0.822 (L21/28)**	**0.540**	cliff

The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking ("repeat-then-judge").

Table 6: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes}: text $\to$ audio	audio $p(\text{True})$-pre AUC
MiniCPM-o 4.5	duplex	0.055 $\to$ **0.556** (collapse)	0.786 (trap dead)
MiniCPM-o 2.6	duplex	0.196 $\to$ 0.345	**0.491** $\approx$ chance
Qwen2.5-Omni	omni-streaming	0.279 $\to$ 0.213 (intact)	0.727
Freeze-Omni	frozen + adapter	0.165 $\to$ 0.112 (intact)	0.612 (capability caveat)

Probe $\oplus$ repeat-then-judge fusion. The restored self-evaluation is additive with the probe. We stack the deployed L22 probe with $p(\text{True})$-pre asked over the model's own transcript (one short text prefill + a 1-token decode per turn; logistic stacker fit on the 360 calibration rows, native regime, Table 7). On the internal test split the fusion is the first significant AUC lift over the deployed probe (0.836→0.862; bootstrap 95% CI on Δ excludes zero), and it transfers to one external pool (TriviaQA 0.789→0.820); the remaining English pools move positively but within CI, and the Mandarin reasoning pool is untouched: there the self-evaluation itself carries no signal (AUC 0.586). Contrary to our a-priori expectation, the gain tracks where the self-evaluation signal is strong, not where the probe is weak; a diagnostic arm asking $p(\text{True})$-pre on the original query text scores within 0.01–0.02 of the transcript arm on every pool, so transcription quality is not the limiter.

Table 7: Probe $\oplus$ repeat-then-judge $p(\text{True})$-pre fusion (native regime, deployed read point). Δ = fusion $-$ probe, bootstrap 95% CI.

pool	n	probe	rtj solo	fusion	Δ [95% CI]
internal test	240	0.836	0.836	**0.862**	$+0.026$ $[+0.004, +0.048]$
TriviaQA	250	0.789	0.761	**0.820**	$+0.030$ $[+0.000, +0.062]$
WebQ	250	0.745	0.723	0.771	$+0.026$ $[-0.007, +0.059]$
Llama-Q	250	0.686	0.674	0.691	$+0.006$ $[-0.020, +0.031]$
SD-QA	200	0.803	0.742	0.808	$+0.004$ $[-0.023, +0.032]$
Reasoning-zh	202	0.640	0.586	0.618	-0.021 $[-0.076, +0.032]$

Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (0.400→0.450), audio overconfidence (paired p_{yes} shift $+0.089$; failure subset 0.415→0.581), and the layer structure (text-calibrated read on real-speech audio holds 0.76–0.80 at L22–L26, peak 0.800 at L25, transfer across both content and modality).

Synthesis. End-to-end multimodal training builds the shared mid-layer core; duplex-style training additionally damages the readouts; omni-streaming gets both right; an adapter alone gets neither (probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

The deployed probe, and which calibration each result uses. The mechanistic probe is a single-position L22 read (4096-d), fit on the 360-query calibration split; every representational result (§3, Appendices B, D) is this probe. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean; 12,288-d), refit on a calibration mix widened to 2,310 public-benchmark queries disjoint from every evaluation benchmark; a pre-registered guard held (frozen internal test AUC 0.860→0.879) and feature selection never saw the externals. It drives the turn-based live loop (§5.1), the frozen transfer results (§5.2), and the talker-enabled replay, frozen throughout, calibrated on text and speech-rendered internal data only. The serving-regime validations are the exceptions. In the concurrent regime, the same 12,288-d feature set is refit in-regime on 360 rows for the loop of Table 15 and on the full 2,310-row mix to measure the effect of calibration scale (Appendix H). In the native regime, the same feature set is refit at the native listen→speak read point; the deployed native gate is the 5,228-row official-configuration merge of Table 18 (Appendix O).
Selection history. L22 and the three-position feature set were chosen on internal calibration-split cross-validation before any external benchmark was scored (the 8-position $\times$ layer sweep and the 19-configuration feature sweep are both calibration-only), and the internal frozen test split was the pre-registered guard. The external pools were then read once with the frozen artifact. Two honest exposures remain: the layer band was picked on internal LOPO rather than by nested selection inside each held-out pool, and the second duplex family's mid-band peak (L30–34 of 56) was located with its own sweep, so both layer choices are best read as confirmed on held-out pools, not as untouched-holdout estimates. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe's in-distribution area advantage is modest: its real advantage is LOPO robustness, which an in-mix test cannot show.

C Second duplex family: NemotronLabs-VoiceChat-11B

The pre-registered prediction test: with the identical recipe (same audio, same judge, 600-query calibration), the layer sweep on this hybrid-Mamba duplex model (56 blocks: 27 Mamba-2 / 4 attention) peaks mid-band at L30–34 (eot-last AUC 0.714 vs. 0.693/0.682 at the endpoints); stacking the same three positions lifts OOF 0.714→0.761→0.790; the frozen-methodology probe transfers to the external pools at AUC 0.781/0.793/0.754/0.701, the deployed-probe band reached from a 600-query calibration set. Boundary notes: the backbone is much weaker at knowledge retrieval (striviaqa local 0.32 vs. MiniCPM 0.62, same judge) and English-only (the cross-lingual pool has no analog). The live-loop port and the trained-checkpoint ablation remain future work; the port is gated by the released inference path, not by the architecture. The only supported loading path (the NeMo Speech `nemotron-labs-voicechat` branch) runs the duplex loop cacheless: every 80 ms frame re-runs the full prefix, an $O(T^2)$ schedule that cannot hold the frame clock in real time, although the hybrid backbone's Mamba-2 state is exactly the $O(1)$ per-frame recurrence a streaming port would need; wiring stateful inference into the released duplex frame protocol is the engineering gap. (Offline, the same property is a convenience: the final frame's forward contains every position, so one hook captures the whole eot window.) Accuracy-side conclusions do not wait on the port: the re-mix arithmetic below tracks measured live arms at 0.011 mean deviation on MiniCPM, inside the $\pm$0.02–0.03 replication floor.

Escalation re-mix on every transfer pool. Completing the grid of Table 1 on this family: an offline re-mix (the same top-r reconstruction arithmetic that, on MiniCPM, tracks the measured live arms at 0.011 mean deviation) sends the top-r of each pool by NVDA probe score to the measured GPT-5.5 outcome and keeps NVDA's own answer for the rest, scored with each pool's official protocol (the OAB judge for the three OpenAudioBench pools, NVDA local floors 0.352/0.376/0.705; our judge for SD-QA, VoiceBench 1–5 for AlpacaEval; the expert answers the heard transcript and is judged directly, so the relay-back channel is excluded; including the measured relay outcomes where they exist shifts tier points by $\leq$0.032). **Selective escalation beats matched-rate random on all five pools** (permutation $p \leq 0.0004$ at the 30% and 50% budgets, $p < 0.03$ at 15%; Table 8, Figure 5), including Web Questions, SD-QA and AlpacaEval, the three pools that are flat or negative on MiniCPM (§5.2). Selectivity is nearly unchanged (AUC 0.72–0.78 under official labels); what changed is headroom (floors of 0.31–0.38 against MiniCPM's 0.51–0.66) and on AlpacaEval the failure species: the terse English-only model fails open-ended queries hard (the probe-selected half scores 3.01 vs. 4.08 kept), where MiniCPM's open-ended failures are soft quality gradations a wrongness detector cannot see. The negatives are properties of the pool $\times$ backbone pairing, not of the signal.

Table 8: The frozen-recipe NVDA probe re-mixed at fixed escalation rates, official judge per pool, each against its own matched-rate random reference (same measured outcomes, escalation set drawn at random; identical at 0% and 100% by construction). AlpacaEval rows are VoiceBench 1–5 scores; others accuracy. Every pool clears random at all three budgets: $p \leq 0.0004$ at 30% and 50%, $p < 0.03$ at 15% (20k permutations, `scripts/18_nvda_random_check.py`).

pool	judge	n	0%	15%	30%	50%	100%
Speech TriviaQA	OAB	247	0.356	0.498	0.615	0.741	0.955
matched random				0.446	0.536	0.656	
Speech Web Questions	OAB	250	0.376	0.480	0.588	0.696	0.828
matched random				0.444	0.512	0.602	
Llama Questions	OAB	234	0.705	0.786	0.838	0.876	0.932
matched random				0.739	0.773	0.818	
SD-QA	ours	200	0.310	0.425	0.545	0.675	0.890
matched random				0.397	0.484	0.600	
AlpacaEval (1–5)	VB	199	3.55	3.83	4.13	4.46	4.92
matched random				3.75	3.96	4.23	

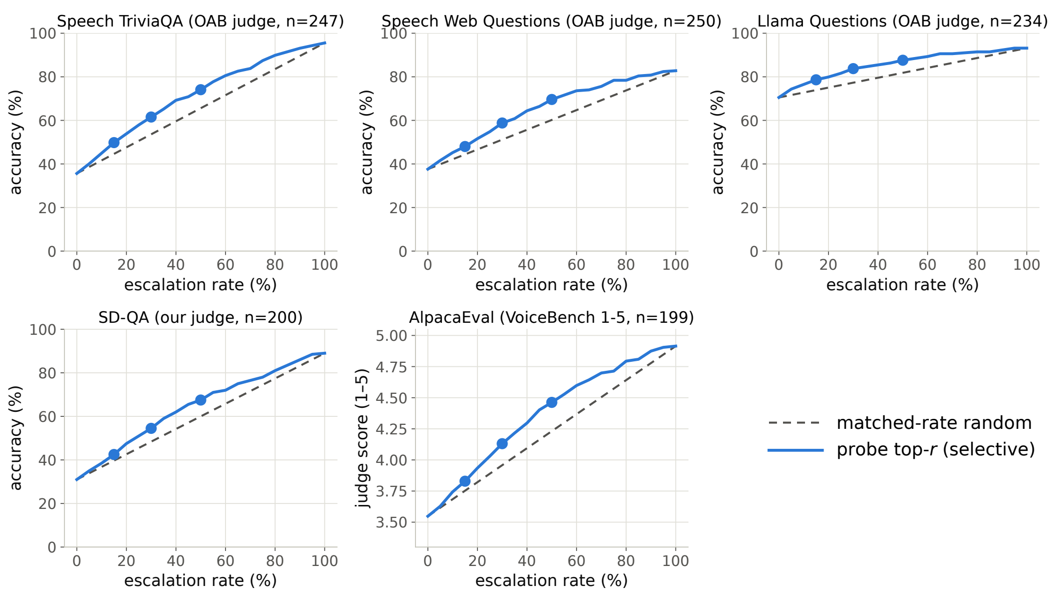
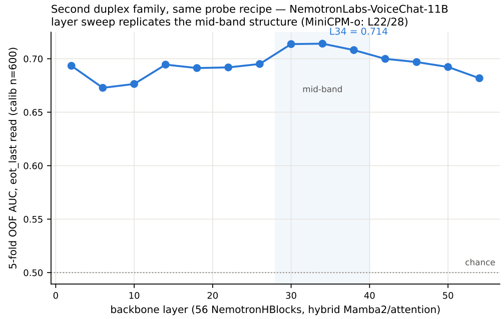
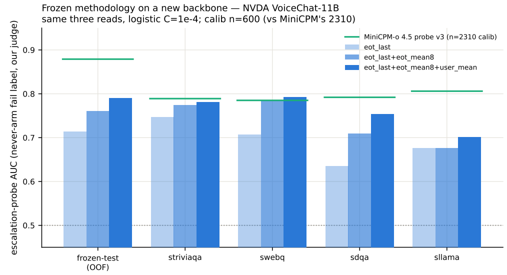


Figure 5: Re-mix curves for the frozen-recipe NVDA probe: selective (top-r by probe score) vs. matched-rate random on the five transfer pools, official judge per pool. Every pool separates, including the three that are flat on MiniCPM (all five pools, $p \leq 0.0004$ at 30/50% and $p < 0.03$ at 15%, permutation).

Calibration expansion, read point, and the price of a causal read. Expanding the NVDA calibration from 600 to 2,481 committed English queries (the same frozen/expansion/expansion2 mix as MiniCPM, no-commit turns dropped) and moving the read from the end of user audio to the model's own commit-to-speak moment — the first run of three non-PAD agent tokens, read over the following eight 80 ms frames, the analogue of the MiniCPM native read (§H) — lifts calibration OOF to 0.820 and cold external AUC to 0.838/0.851/0.771/0.774 (mean 0.808; Table 9). The reported metrics use the full 2,481-row analysis cohort. The exported deployed, v2, and causal artifacts are



(a) layer sweep



(b) external transfer

Figure 6: Second-family test: (a) the usable read again peaks mid-network; (b) frozen-recipe transfer lands in the deployed-probe AUC band.

separate refits that exclude the frozen 240-row test split before dropping no-commit turns, leaving 2,258 calibration rows; those artifacts therefore do not share the table fits. The layer choice is flat: L26–L34 are within 0.005 on every read. Two facts about this read point were established by a third, independent replay of every query (Figure 7a). First, the read is not pre-answer: the deployed window covers the model's first ≈640 ms of answer, and a *strictly causal* probe that reads nothing after the commit frame (commit-frame state, the eight frames before it, and the running mean of the user audio up to the commit) costs 0.025–0.030 external AUC at every layer (0.781 at L34, 0.785 at L30, paired bootstrap CIs exclude zero). The pre-commit window is no better than the end-of-audio window (0.782), so the extra signal is the answer onset itself, not anything heard before the commit. Second, the online readout's running user-audio mean (which cannot know where the audio will end) is interchangeable with the offline mean over the full utterance: cosine 0.996 median, -0.004 $[-0.009, +0.001]$ external AUC. The recipe itself is at its ceiling: ≈85 variants proposed by two independent reviewers (alternative linear heads, shrinkage LDA, per-frame and multi-layer features, re-weighting, robust scaling, PLS/PCA) move external AUC by at most $+0.013$, and the best calibration-selected variant — a three-layer (L26/L30/L34) average of linear heads with the commit frame added as a fourth block, $+0.004$ to $+0.007$ AUC, replicated on the third replay — delivers the same accuracy as the deployed read recipe at every escalation budget (Figure 7b; both beat matched-rate random on all four pools at 15/30/50%, $p \leq 0.065$, $p \leq 0.002$, $p < 0.001$). What does move: as on MiniCPM, fixed calibration thresholds do not deliver the nominal budget across pools (the 15/30/50% tiers fire on 1/4/12% of Llama Questions and 7/27/53% of SD-QA), so per-pool quantile thresholds are the deployable operating point.

Table 9: NVDA probe read points (linear head, L2-logistic C=10^{-4} on standardized features, full 2,481-row analysis fits; externals cold, $n = 249/250/249/200$). $\text{LOPO}_{\text{pool}}$ = mean within-held-pool AUC under leave-one-calibration-pool-out (13 eligible pools). Δ = paired-bootstrap difference in external mean vs. the deployed read on the same replay. The exported artifacts are separate 2,258-row refits that exclude the frozen test split.

read	features	OOF	$\text{LOPO}_{\text{pool}}$	external mean	Δ [95% CI]
end of user audio, L34	last ‖ mean_8 ‖ user-mean	0.810	0.680	0.782	-0.027 $[-0.049, -0$
strictly causal, L34	commit frame ‖ pre-mean_8 ‖ running mean	0.812	0.694	0.781	-0.029 $[-0.048, -0$
strictly causal, L30	same	0.805	0.678	0.785	-0.025 $[-0.044, -0$
deployed read recipe: commit + 8 frames, L30	onset-last ‖ onset-mean_8 ‖ user-mean	0.816	0.685	0.809	—
deployed read, causal user-mean	onset-last ‖ onset-mean_8 ‖ running mean	0.814	0.684	0.805	-0.004 $[-0.009, +0$
v2: 3-layer average, L26/30/34	+ commit frame, causal user-mean	0.818	0.691	0.813	$+0.004$ $[-0.005, +0$

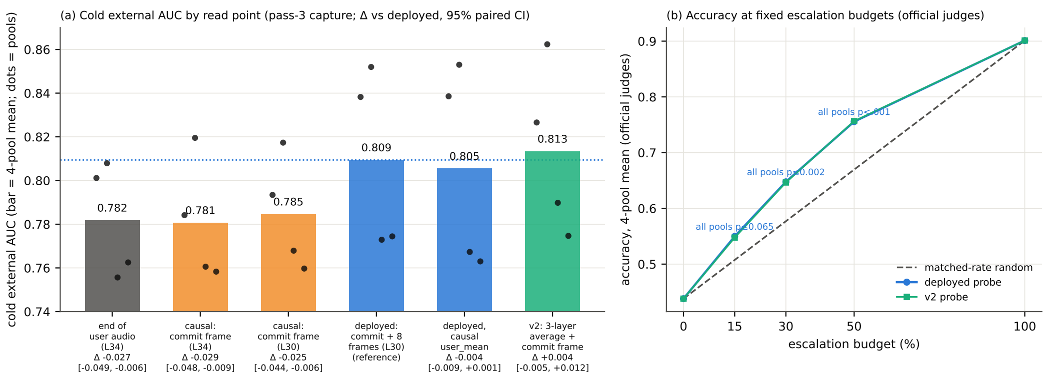


Figure 7: NVDA read points. (a) Cold external AUC by read on the third replay: a strictly causal read (orange) costs ≈0.03 AUC relative to the deployed commit+8-frame read (blue); the reviewed v2 recipe (green) adds $+0.004$. Dots are the four pools; Δ is the paired-bootstrap difference vs. deployed. (b) Accuracy at fixed escalation budgets, four-pool mean under the official judges: the deployed and v2 probes are indistinguishable, and both beat matched-rate random at every budget.

D What breaks at the output: mechanism audit

§3.2 reports that the final-layer last-token read inverts. This section reports what we could and could not establish about why. Everything below uses the Phase-5d all-layer prefill dumps (every decoder layer, last-token and mean-pooled, same 600 queries) and the identical probe protocol as the layer sweep: calibration split, L_2 logistic regression, LOPO trained on the four non-math pools and scored on held-out math. The reproduction check recovers the published L22 and final-layer numbers to $\leq$0.012 (`scripts/14-17_*.py`).

Four mechanisms ruled out. Table 10 lists them. (i) **Speak mode.** Prefilling the same text queries under the TTS/speak-mode template leaves the cliff exactly where it was (final-layer LOPO math 0.362 vs. 0.366 plain), so the readout is not being displaced by a prepare-to-speak state. (ii) **Modality axis.** The mean audio-minus-text direction at the final layer is orthogonal to both the inverting final-layer probe and the transferring L22 probe (cosine 0.013 and 0.027; chance is $1/\sqrt{4096} = 0.016$), so the cliff is not the audio/text split that §3.3 measures. (iii) **More shortcut available late.** Source-pool identity is linearly decodable at 0.88–0.96 at every depth, in the duplex model and its raw backbone alike: the query-type shortcut of §3.1 is not something the late layers add; it is what remains once the competence signal stops being readable. (iv) **Wholesale representational drift.** Raw linear CKA between the duplex model and its backbone appears to collapse late (0.80→0.47), but transformer residual streams carry a few massive-activation coordinates that dominate the Gram matrix; with per-feature standardization the same comparison is nearly flat (0.82→0.78) and does not localize the cliff. Projecting out the massive-activation axes, or the pool-mean directions, rescues nothing (0.35–0.40 at every $k \leq 20$, against a random-direction control of 0.37).

Table 10: Mechanism candidates for the final-layer inversion. Final-layer LOPO hard-math AUC on MiniCPM-o 4.5 unless noted; 0.366 is the unmodified baseline and 0.931 the L22 read.

candidate	test	verdict
prepare-to-speak state	speak-mode template prefill	0.362, no change
modality (audio vs. text) axis	cosine with the probe direction	0.013 ≈ chance
late layers encode more pool identity	pool decoding by depth	0.88–0.96 at all depths
representation rewritten late	standardized CKA vs. backbone	0.82→0.78, flat
massive-activation coordinates	project out top-k such axes	0.35–0.40, no rescue
query-type directions	project out pool-mean directions	0.34–0.45, no rescue
delivery to the output position	**residual component of the read**	**0.93→0.27 (Fig. 8)**

What does separate the models. Figure 8 measures the probe rather than operating on the model. For the probe trained at each depth on the four non-math pools, we split its held-out score into the part carried by that layer's five dominant principal directions (estimated from training rows only) and the residual, $s = w^\top Px + w^\top(I - P)x$. The residual component is the distributed read, and it is the one that transfers: it holds 0.85–0.93 through mid-network on MiniCPM-o 4.5 and then decays to **0.27** at the output, with the same decay on MiniCPM-o 2.6 (0.81→0.52) and no decay on either raw backbone (Qwen3-8B 0.86, Qwen2.5-7B 0.71 at their final layers) or on the omni-streaming control (Qwen2.5-Omni holds 0.75 to its output with a flat variance share): the decay tracks duplex fine-tuning specifically, not streaming or audio fine-tuning in general. On MiniCPM-o 4.5 the score also concentrates: the dominant subspace carries 39% of the held-out score variance at the final layer against $\leq$0.18 at every earlier depth and 0.05 for its backbone; on MiniCPM-o 2.6 this concentration is present but weak (0.16 vs. 0.14 for its backbone), so we report it as a property of the stronger duplex model, not a general law.

Direct turn-control tests. Two experiments target the turn-control story head-on rather than by exclusion (`modal_interp.py`, `scripts/19_*.py`). **Control-token logit lens:** applying each model's own final norm and unembedding to the stored per-layer states, the probability mass on the trained-in control vocabulary (added/special tokens) rises over MiniCPM-o 4.5's last layers (log-mass -23.8 at L32 to -11.8 at the output) while the raw backbone's falls by fifteen log-units over the same span: the duplex output layer, uniquely, stays ready to emit control tokens. But the fine-grained prediction fails: the inverting probe direction is no more aligned with the control tokens' unembedding rows than with random vocabulary rows (max $|\cos|$ 0.069 vs. 0.068), and per-query control mass correlates with the probe score at only r=0.18. **Listen/speak intervention:** re-prefilling 60 calibration queries with a still-listening vs. answer-now suffix moves the final-layer state substantially in both models (relative displacement 0.27 duplex, 0.59 raw; cue text is just prompt semantics), and the cue direction is orthogonal to the inverting probe ($|\cos|$=0.019, chance 0.016), shifting its score by only $+0.23\,\mathrm{sd}$ (and the deployed L22 read by $+0.08\,\mathrm{sd}$). Together: the duplex output layer is measurably control-biased in its output distribution, but the inversion is not carried by control-token directions or by an explicit listen/speak axis: the simple takeover story fails its own direct tests.

What we do not claim. Destructive tests do not settle the mechanism. Removing the top five principal directions from the final layer does raise LOPO math from 0.378 to 0.758, but the identical operation costs the raw backbone 0.90→0.14 and costs the intact L22 read 0.931→0.760, so it is a damaging intervention that happens to help a broken readout, not evidence of a duplex-added subspace. We therefore state the finding at the level the evidence supports: duplex fine-tuning does not erase the competence signal, it stops delivering it in distributed form to the final last-token position, and the layer where delivery fails is also the one whose output distribution stays control-biased. Attributing the loss to a specific duplex objective needs training-time access we do not have; the practical consequence, read mid-network, does not depend on the attribution.

E Query-surface router audit

A RouteLLM-style TF-IDF→LR router under identical labels reaches OOF AUC 0.669 (below the pool oracle's 0.715); a 24-configuration sweep tops out at 0.689, so this is the query surface's ceiling, not tuning. Under LOPO it collapses to 0.38–0.57 and scores the 100%-fail trap pool at 0.230: it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 10). Nor is it data-starved: the identical recipe on RouterBench [Hu et al., 2024] (36,497 prompts) reaches only in-domain AUC 0.710 with leave-one-benchmark-out at chance; RouteLLM's released preference-trained routers score 0.523/0.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (0.440). On audio labels the router improves (test 0.814) but still trails the mid-layer probe (0.879) and cannot fire mid-stream. In distribution, the gate's area over random ($+0.054$) is statistically on

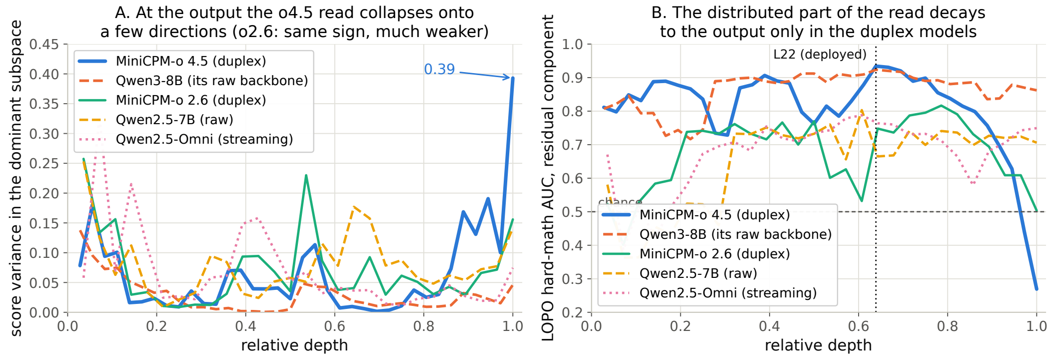


Figure 8: Non-destructive decomposition of the probe's held-out score at every depth. **Left:** share of score variance carried by the layer's five dominant directions. **Right:** transfer AUC of the residual (distributed) component, intact to the output on both raw backbones and on the omni-streaming control, decaying to inversion over the final layers of both duplex models.

par with the pool oracle's ($+0.042$; paired bootstrap delta CI $[-0.003, +0.027]$, p=0.13): the gate's case is transfer, not the in-distribution margin. $p(\mathrm{True})$ tradeoff variants: Figure 11.

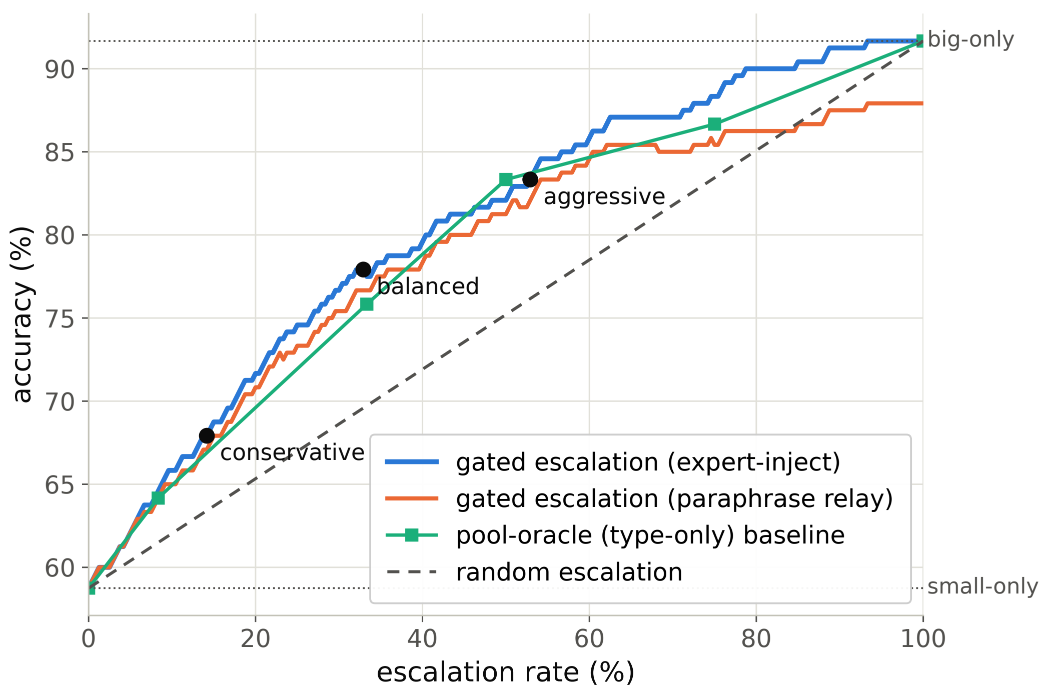


Figure 9: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

F Gate timing details

Against the model's own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode, self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency at every escalation rate (gated-cloud@33%: 78.7% at mean 6.5 s vs. all-THINK 63.7% at 22.4 s; Table 11). The reason is mechanistic: the gate flags knowledge and trap failures, which extra compute cannot fix, so external escalation is necessary, not merely better.

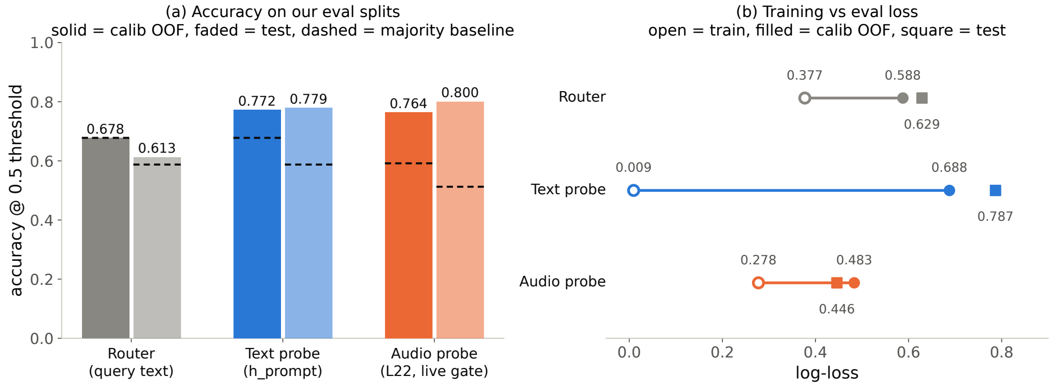


Figure 10: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

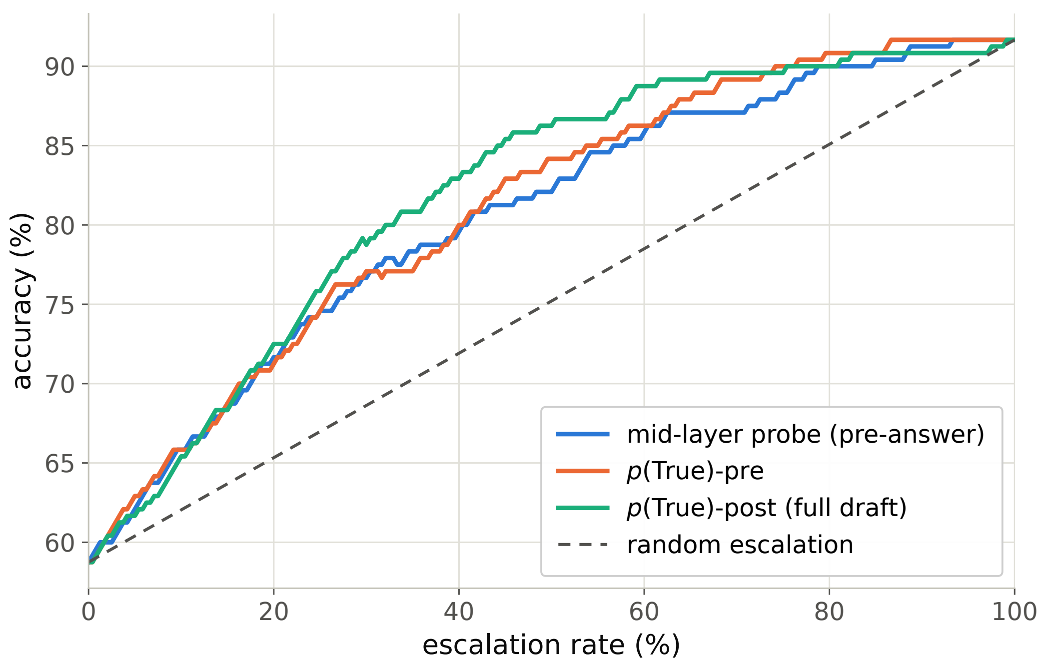


Figure 11: Offline tradeoff for the $p(\mathrm{True})$ signals (pre-answer and post-draft).

Measurement protocol. Gate latency is wall-clock from prefill start to the L22 score being available (truncated forward + probe dot product), CUDA-synchronized, one H100, 50 queries per modality, three warmup calls excluded, interleaved per query with the reference arms; P50/P95: gate 20/25 ms text and 45/104 ms audio vs. full-prefill TTFT 36/47 and 68/144 ms. Network and expert-RPC initiation are excluded: the cloud call fires asynchronously at the gate, and its user-facing cost is reported as expert-content-audible latency in Appendix G. Absolute prefill wall-times vary with call overhead across benches; the robust cross-bench statistic is the ratio: the L22 state is ready at $\sim$57–61% of prefill.

Per-layer truncated-forward cost vs. quality (Figure 12): the L22 hidden state is available at $\sim$57–61% of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at $\sim$55–60% depth. If the cloud call launches at the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but

Table 11: Deployment policies, frozen test split (n=240); latencies measured.

policy	acc (%)	lat. mean (s)	P50	P95
fast-only	58.8	3.0	3.5	4.2
all-THINK (built-in)	63.7	22.4	17.2	60.1
gated-think @33%	61.3	12.1	3.6	47.6
gated-cloud @15%	68.8	5.3	3.6	10.1
gated-cloud @33%	**78.7**	**6.5**	3.6	20.8
gated-cloud @50%	85.8	7.0	3.6	24.4

only ∼20–31% of escalated queries (they skew to short local answers); the residual wait is P50 2–3 s, P90 ∼27 s (Figure 13), the stall-phrase budget the injection protocol must cover.

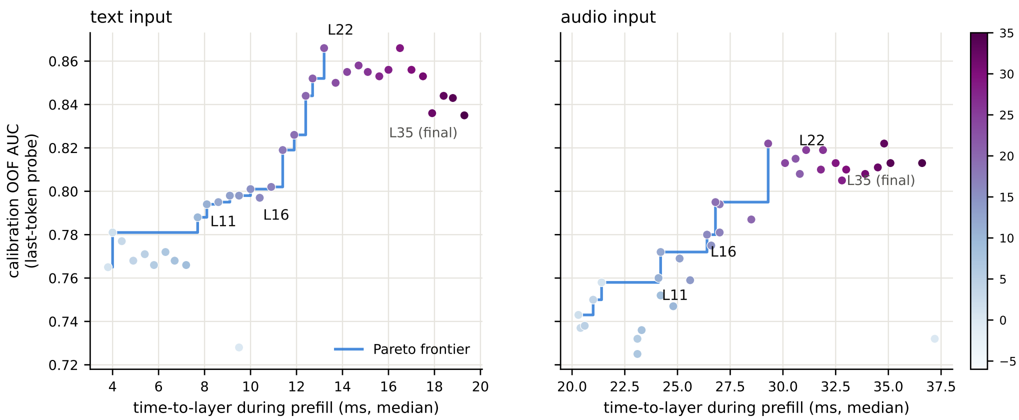


Figure 12: Fork-depth Pareto: per-layer wall-time vs. gate quality.

G Live-loop details

What the signal detects. The pre-answer probe is specifically a retrieval-failure detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is observable in the model's state. Execution failures (stuck mid-derivation) surface in decode-time signals instead, and metacognitive failures (false-premise questions, where no failure event occurs at all) are invisible to every signal we tested, query-surface routers included (escalation rate 18% vs. trap's 90%; Appendix G), which motivates a decode-time check behind the pre-answer gate.

Full live tables. Table 12 (both scoring views, speakable subset and full pool) and Table 13 (measured response latency). The conservative tier shortens the P95 tail: its few escalations displace exactly the longest local decodes. Figure 14 joins accuracy and latency; Figure 15 replays three real sessions from recorded timers.

Time to first audible audio. The sweep above is text-out; a TTS-on re-run of the balanced arm (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker's own voice) measures speech onset directly. The gate decision is unchanged: the TTS token enters the context only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From gate decision to first audible audio: **local answers p50 0.552 s / p95 0.599 / p99 0.644**; **escalated turns p50 0.022 s / p99 0.029**: the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the escalated path reaches first audio 25× faster. Expert content becomes audible at p50 6.4 s / p95 26.0 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 0.66 s after the answer returns); per-turn dead air between stall end and expert content is p50 2.5 s / p95 22.4 / p99 30.7, with 25/84 escalated turns fully covered; the stall covers the head of that wait, and Table 13's completion profile bounds the tail.

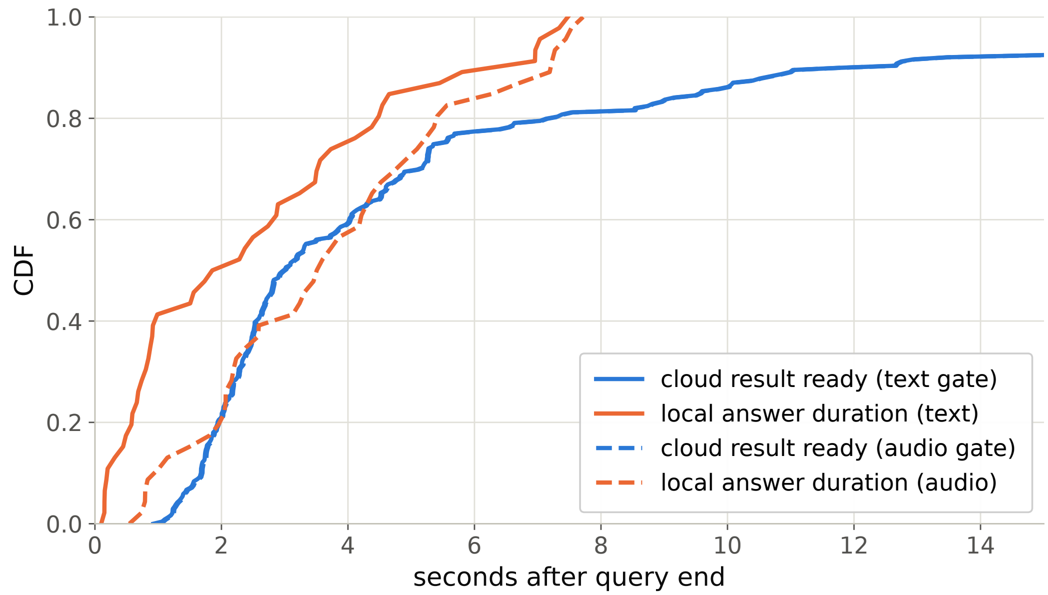


Figure 13: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

Table 12: Live streaming sweep, paired bootstrap 95% CIs (B=10^4). Top: speakable subset (n=218); bottom: full pool (n=240). Gold-inject = escalated rows re-scored with gold-text expert answers; channel cost = paired difference. * CI excludes 0; † CI lower bound at zero.

tier	esc	deployed	Δ vs. floor	gold-inject	channel cost
never	0%	0.422 [0.36,0.49]	—	0.422	—
conservative	14%	0.528 [0.46,0.59]	+0.106*	0.550	+0.023†
balanced	31%	0.592 [0.53,0.66]	+0.170*	0.661	+0.069*
aggressive	57%	0.633 [0.57,0.70]	+0.211*	0.761	+0.128*
always (measured)	100%	0.665 [0.60,0.73]	+0.243	0.922 [0.89,0.95]	+0.257
never (full)	0%	0.383 [0.32,0.45]	—	0.383	—
conservative	16%	0.483 [0.42,0.55]	+0.100*	0.525	+0.042*
balanced	35%	0.554 [0.49,0.62]	+0.171*	0.654	+0.100*
aggressive	61%	0.596 [0.53,0.66]	+0.212*	0.771	+0.175*
always (measured)	100%	0.617	+0.234	0.917 [0.88,0.95]	+0.300

Speculative prefetch, measured and dropped. Framed as speculative escalation (draft–verify at sentence granularity), its acceptance rate (does an expert answer to the partial transcript answer the full question) is 0.07/0.19/0.51 at 25/50/75% of the audio; at the mid-stream gate's typical fire point 6–8 of 10 calls would be wasted. The trap pool's floor (0.00/0.10/0.27) shares the back-loaded-core property that breaks mid-stream probe reads.

Conflicting-injection controls (172 sessions). Fabricated conflicting expert answers by within-pool derangement, four framings: worst-case comply is 0.00 neutral / 0.53 deployed authority / 0.79 strong-override, while seeding words into the talker's mouth backfires (comply 0.25, lip-service 0.62; forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed. Model-wrong $\times$ wrong-injection swallow rate is 0.64: expert quality is the accuracy ceiling of the cascade.

Uplink diagnostics. On the escalated population the expert answering the talker's own transcript scores 0.585 vs. 0.871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights

Table 13: Measured response latency (s), n=240 per arm (latency is channel-independent). P99 on n=240 is indicative only.

tier	mean	P50	P95	P99	Δmean	ΔP50	ΔP95	ΔP99
never	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

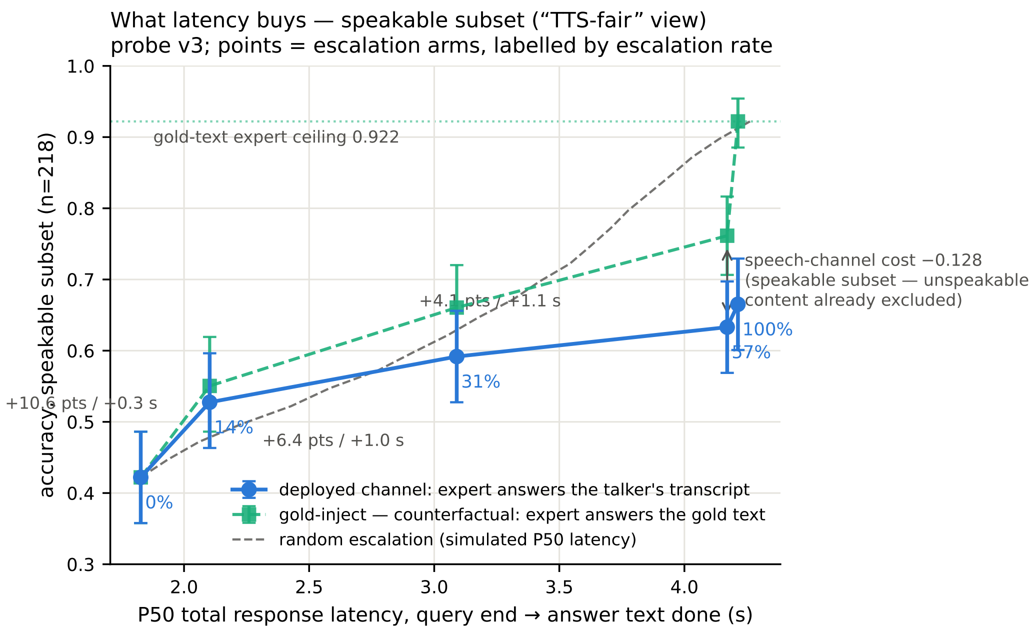


Figure 14: Accuracy vs. per-arm P50 latency (speakable subset), both scoring views, with the simulated-P50 random reference.

Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap (0.585→0.694, McNemar p=0.007), diagnostic only (its critical-path latency and external-pool survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the audio channel itself is lossless for speakable content: the leak is the talker's transcription, not the synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated failures, first-generation sweep).

Boundary query classes. **False premises**: the model fails substantially (0.63 text / 0.47 audio) and the expert would help (0.80), but the gate is blind (fire@balanced 0.18 vs. trap's 0.90; no score separation): answering along a false premise does not feel like inability. Real-time queries ("what is NVDA trading at"): the gate is not blind, the class was simply absent from every static training family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori, never-changing as in-family controls) raises held-out fast-changing escalation rates to 0.80/0.93/1.0 across tiers with frozen-test AUC unchanged (0.877 vs. 0.879). One calibration subtlety: tier budgets must remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold shift.

Demonstration system. The pipeline runs as an interactive web demo on one H100, on the model's native full-duplex head (Appendix O): every second of microphone audio is prefilled into the context the talker is generating from, the head itself decides listen/speak per chunk, and the gate reads L22 at the listen→speak transition against the in-regime thresholds. Escalated turns play a canned stall

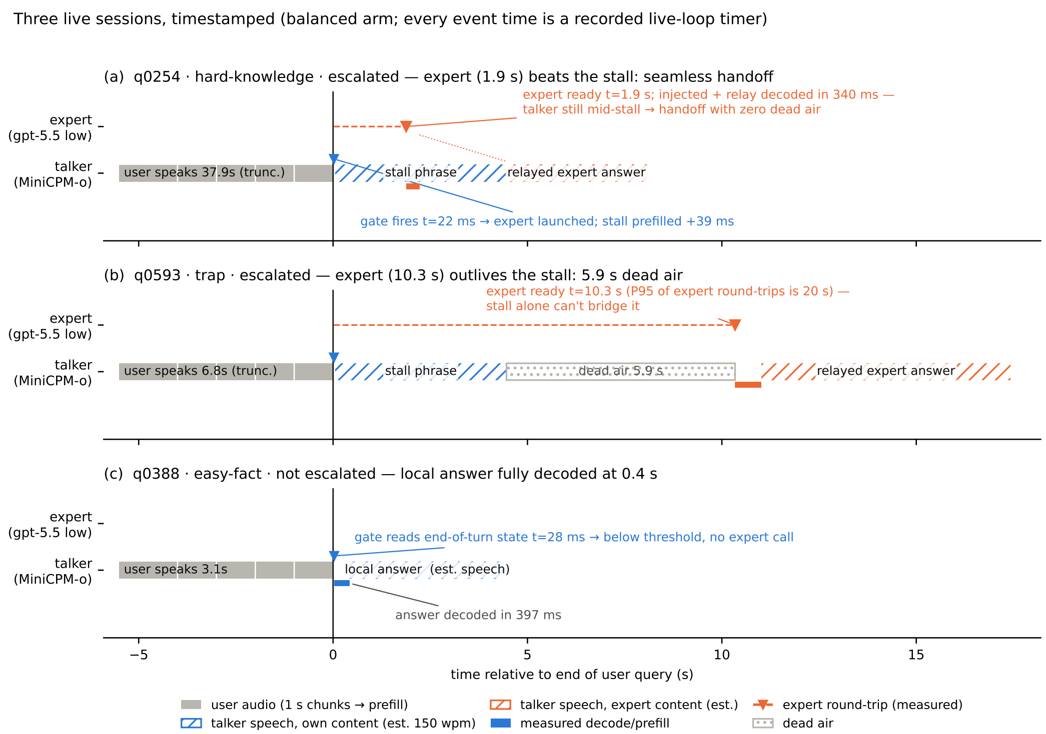


Figure 15: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

(teacher-forced once, the talker's own voice), consult the expert (with a demo-only web-search tool for real-time queries), and the verified answer re-enters the stream as a text unit that the talker voices, an ordinary duplex turn, hence interruptible like any other. There is no VAD, no ASR barge-in classifier, and no abort machinery: floor control is entirely the duplex head's own behavior (Appendix O). An earlier demo build approximated barge-in with an energy-VAD + burst-ASR harness over the turn-based loop; it is retired, and survives only as the harness-overlap arm measured above. The demo will be made available upon publication.

H Duplex-validation sweep: the gate under the talker

The talker-disabled live sweep (§5.1) never engages the talker head; this sweep replays its full 240-query pool through the deployed voice loop above to test whether the frozen probe read survives the spoken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation disabled (the arm isolates gate validity, not routing outcomes). **Clean**: the query streams as mic-shaped PCM (pool TTS audio plus steady noise at RMS 0.008), the energy VAD ends the turn, the talker answers aloud, and the session is cut after the first audible chunk. **Overlap**: an easy warmup question (probe score $<$0.25, locally answered, $>$5 s spoken answer) puts the talker mid-answer, and the target query is then spoken over it; sustained overlapping speech commits a barge-in whose audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

Against the frozen local-failure labels of the same pool, the identical frozen probe reads: talker-disabled AUC 0.866 $[0.817, 0.909]$, clean 0.871 $[0.824, 0.914]$, overlap 0.856 $[0.803, 0.901]$; barged-only subset (n=237) 0.854. Paired bootstrap deltas against the talker-disabled arm are $+0.005$ $[-0.025, +0.035]$ (clean) and -0.009 $[-0.036, +0.016]$ (overlap), both inside the $\pm$0.02–0.03 live replication floor. Per-query scores correlate across regimes at r=0.90–0.92, and the balanced threshold makes the same escalate-or-hold decision as the talker-disabled read on 0.887 (clean) and 0.912 (overlap) of queries. The end-of-turn read costs 24–26 ms at p50 (p95 $\leq$30 ms) in both arms, with the first audible answer chunk arriving $\geq$0.5 s after the gate decision (p50 $\approx$0.58 s), so the decision precedes any audible commitment. Scope: this loop keeps per-turn session control: the mic stays open while the talker speaks and interruptions seed the next read, but incoming audio is not prefilled during generation. In hindsight this sweep and the concurrent arm are best read as read-point ablations: the deployed native full-duplex regime (Appendix O) sits strictly between the turn-based ideal and the harness-concurrent worst case on every pool measured. The concurrent regime is measured separately below.

Floor control under escalation: barge-in vs. backchannel. The demonstration loop's floor policy (speech over the talker ducks playback; a fast ASR pass returns the floor on backchannels or commits the interrupt) raises a distinct worry: did adding the gate change the talker's overlap behavior? A dedicated sweep injects overlap stimuli into the deployed loop at real-time pacing, four classes: short backchannels (0.40–0.46 s, the English+Chinese lexicon), long lexicon-only continuers (time-stretched to 1.33–2.08 s, probing the sustained-speech commit rule), out-of-lexicon stop commands, and pool queries spoken over the answer, in matched pairs with the gate on and off under an identical floor policy (non-firing queries, n=40 per class per arm). The answer is exact orthogonality: all 160 pairs make the identical duck/resume/interrupt decision in both arms (every paired rate delta 0.000 $[0.000, 0.000]$; paired latency medians within 31 ms), down to the single seeding failure, which occurs on the same query in both arms. By construction the gate reads once at end of turn and never enters the floor state machine; this measures it. The three phases that exist only under escalation (the stall sentence, the expert wait, the relay speech) are the actual new risk surface and behave (Table 14): a genuine barge-in aborts the escalated turn and seeds the next turn 40/40, and short backchannels hold the floor at the same rate as during local answers. One asymmetry is disclosed: a false interrupt during the expert wait cancels the pending expert call, so the policy's fail-toward-interrupt bias is recoverable but not free there. That residual backchannel false-interrupt rate (0.40 short, 0.50 long; identical in both arms) is a property of the harness policy and is per-stimulus deterministic: non-lexical hums ("mm-hm" and its Chinese counterpart) die by the ASR-empty fail-closed rule (16/16 each; worded tokens such as "okay" 0/16), and continuers sustaining loud speech past the $\geq$1.2 s commit rule die without any ASR check (20/20; pause-bearing variants 0/20). We keep the policy as deployed for measurement integrity and note both rules as the obvious fixes. For scope: the checkpoint's native duplex profile, measured separately on Full-Duplex-Bench v1.0 [Lin et al., 2025] (727 samples, official scripts), shows best-in-class pause handling (turn-over rate 0.125/0.117 Candor/synthetic), competitive user-interruption response (0.915 at 0.90 s), and zero spontaneous backchannels: the lexicon floor is a harness-level approximation added precisely because the talker head has no native backchannel behavior to preserve.

overlap injected during	backchannel holds floor	barge-in aborts	seeds next turn
local answer (gate off)	0.60 $[0.45, 0.75]$	40/40	39/40
local answer (gate on)	0.60 $[0.45, 0.75]$	40/40	39/40
stall sentence	0.63 $[0.38, 0.88]$	16/16	16/16
expert wait	0.60 $[0.33, 0.87]$	14/14	14/14
relay speech	0.55 $[0.27, 0.82]$	10/10	10/10

Table 14: Floor-control sweep over the deployed voice loop. Short backchannels (lexicon bursts) vs. genuine barge-ins (pool queries spoken over the talker), by the phase the overlap lands in; the last three phases exist only under escalation. Interrupt latency p50 1.39–1.40 s from overlap onset in every phase; escalated-turn aborts discard the pending expert answer and the barge-in speech becomes the next turn.

Concurrent prefill: interleaved listen/speak in one KV stream. A third arm drives the streaming API in time division: while the talker is mid-answer, each 1 s chunk of the target query is prefilled into the same session between generation chunks, so the end-of-turn read fires while generation is still active (238/240 queries, full interleave coverage). The carrier answer is teacher-forced: under sampling the duplex model reliably yields the turn: it emits EOS within $\sim$2 s of user audio appearing in its context, the trained turn-taking behavior surfacing under interleave. This is the one shift on the transfer ladder that breaks the frozen read: scores drift up (mean 0.48→0.71; the frozen balanced threshold's escalation rate jumps 37→75%), AUC drops 0.869→0.758 $[0.693, 0.816]$ (paired ΔAUC -0.109 $[-0.158, -0.064]$), and label-free requantiling recovers decision agreement only to 75%. The information, however, survives in the state: refitting the same 12288-d linear read on 360 calibration queries collected in-regime reaches **0.818** $[0.761, 0.871]$ on the held-out 240 (calibration OOF 0.775), more than half the gap recovered from those 360 in-regime rows alone. The mid-layer competence direction persists under concurrent listen/speak; unlike every earlier shift (modality, external pools, talker enabled), calibration must follow the regime. Run-to-run replication of the concurrent read correlates at 0.895. Caveat: the interleave goes through the public streaming API: turn-role headers enter mid-generation and the carrier text is forced, so this measures the concurrent state, not the official duplex serving format.

The full loop in the concurrent regime. With the in-regime gate (thresholds = label-free quantiles of each pool's own concurrent always-local arm scores), the complete escalation loop (carrier speaking, target audio interleaved, EOT read mid-generation, stall → expert → relay) was re-run end to end on all seven pools (Table 15). We report this sweep as exploratory: one session per query per arm, an in-regime probe calibrated on 360 internal rows only, teacher-forced carrier (the caveat above). The loop's mechanics reproduce: realized escalation rates track nominal on the external pools (10.8–51.2%), and latency holds (EOT read p50 23 ms, canned-stall onset 27–34 ms, local first audio p50 .655 s vs. .552 turn-based, the price of the longer concurrent context), and on the internal pool, where calibration matches the regime, so does selection: accuracy climbs 40.4→52.1→66.2% (speakable **44.5→56.0→71.1%**) over the balanced and aggressive tiers, clearing the matched-rate random remix of the same never/always outcomes (p=.017 and 5×10^{-5}; speakable .009 and 5×10^{-5}), and selective escalation beats always-escalate (66.2 vs. 61.7% at two thirds of the calls; speakable 71.1 vs. 66.5): the turn-based signature result carries over. The matched-random rows of Table 15 bound any stronger reading, in three ways. (i) Gains are not monotone everywhere: the internal conservative tier under-fires (2.5% realized vs. 15.8% nominal; the top calibration quantile did not transfer to the regime; balanced and aggressive did) and lands flat (40.4→40.0, 44.5→43.6, inside the $\pm$2–3 floor). (ii) Externally the 360-row in-regime probe separates from matched-rate random only in spots (Llama @50%: 90.8 vs. 86.5, p=3×10^{-4}; Reasoning zh @30/50%: $p \leq .0024$; AlpacaEval @15%: p=.031); at the other external operating points the budget gains are what random escalation would buy, consistent with this probe's external AUC of .57–.67 against the turn-based probe's .68–.81 (internally, where calibration matches the distribution, it reads .857 vs. the turn-based .879). We then tested the calibration-scale attribution directly: the full 2,310-row calibration mix was re-collected in the concurrent state and the same read refit at full scale. Scale is real: every English pool lifts (paired against the 360-row probe on the same features: TriviaQA .645→.699, WebQ .641→.713, Llama Q. .674→.755, SD-QA .639→.701; mean ΔAUC $+.068$ $[.036, .099]$) along a monotone data-scaling curve (external mean .625→.689 over 360→2,310 rows; internal .818 unchanged), but it buys only about a third of the distance to the turn-based .785–.806, and Reasoning zh, absent from the English calibration mix, moves within noise (.615→.577 $[-.117, .037]$). The residue is the regime itself, not calibration scale. Table 15's loop ran with the 360-row probe and its numbers are unchanged. (iii) On Llama Questions selective-vs-always is judge- and noise-sensitive (official judge 90.8 vs. 91.2; ours 86.4 vs. 85.6; both deltas inside the floor), so we claim that comparison only on the internal pool. Floors move by pool, the regime cost story: Llama $\approx$ unchanged, TriviaQA -16, Reasoning zh -18 (the English carrier hurts the zh pool most), WebQ $+4.8$ under the official judge (within the $\pm$2–3 replication floor; the larger our-judge shift is alias-matching style sensitivity). AlpacaEval rises 4.36→4.60 (1–5 scale, always 4.76), but only its conservative tier clears matched random (p=.031; 4.40–4.55 across tiers), so the turn-based honest-negative stands.

A serendipitous recovery behavior. An accidental run surfaced one bonus behavior worth reporting: when a long question contains a $>$1.25 s internal pause, the endpointer fires early and the talker starts answering the fragment, but the question's own continuation then triggers barge-in, and the

Table 15: The full escalation loop in the concurrent regime (exploratory: one session per arm, 360-row in-regime calibration, teacher-forced carrier), all pools, judge-matched to Table 1 (OAB official for TriviaQA/WebQ/Llama; ours for SD-QA/Reasoning/internal; VoiceBench 1–5 for AlpacaEval). Gate = in-regime probe, per-pool label-free quantile thresholds; tier columns are headed by each pool's realized rate, and the *rand* row under each pool is the matched-rate random remix of the same never/always outcomes (20k draws; p = P(random $\geq$ gate), marked $^{*}p<.05$, $^{**}p<.01$). Accuracies in %; AUC = in-regime probe vs. always-local arm failure, decimals. Script: `scripts/20_conclive_random_check.py`.

pool	never	cons.	bal.	aggr.	always	AUC
TriviaQA	50.4	54.0	64.0	71.6	91.6	0.636
rand (10.8/30.8/46.8%)		54.9	63.1	69.6		
WebQ	61.6	65.2	68.8	71.2	81.6	0.636
rand (15.6/30.4/48.8%)		64.7	67.7	71.4		
Llama Q.	81.6	83.2	84.8	90.8**	91.2	0.672
rand (14.8/29.2/51.2%)		83.0	84.4	86.5		
SD-QA	49.5	55.5	60.5	70.5	89.5	0.636
rand (15.0/29.5/50.5%)		55.5	61.3	69.7		
Reason. zh	40.6	48.5	58.4**	64.4**	80.7	0.570
rand (15.3/30.2/45.0%)		46.8	52.7	58.7		
internal (240)	40.4	40.0	52.1*	66.2**	61.7	—
rand (2.5/36.2/66.2%)		41.0	48.1	54.5		
internal speakable (218)	44.5	43.6	56.0**	**71.1****	66.5	0.857
rand (2.3/31.2/62.8%)		45.0	51.4	58.3		
AlpacaEval (1–5)	4.36	4.45*	4.53	4.60	4.76	—
rand (10.1/30.7/47.7%)		4.40	4.48	4.55		

remainder becomes the next turn, where the gate simply re-scores it. Early end-pointing thus degrades into an extra turn rather than a lost query, the honest duplex-ish recovery available to a turn-based loop.

I Fixed-threshold transfer and the expert-benefit oracle

Both analyses below run in the gold-inject view (escalated rows take the expert's answer to the gold text). The deployed-channel view cannot evaluate a freely moving threshold: the most permissive arm escalates 50%, so the other half of each pool has no expert outcome on record, whereas the ceiling runs cover every id. Labels follow each pool's own judge (Table 1); the two judges disagree by up to 13 points on Web Questions, so they are never mixed within a row. AUC columns are recomputed from the deployed sweep's stored end-of-turn scores; they differ from Table 1's AUC column (captured in a separate scoring pass) by up to 0.03 externally and 0.04 on the internal pool (0.840 here vs. 0.879 there): the passes render and prefill the audio independently, and the probe score carries render-to-render jitter. AlpacaEval is excluded: it has no binary correctness label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

One absolute threshold, six pools. The deployed gate adapts per pool, but only through label-free quantiles of its own score distribution. To separate what the probe's ranking carries from what its scale carries, we freeze the single absolute threshold the internal system actually deployed (τ=0.628, the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer**: the same τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a fixed cut cannot hold an escalation budget across pools. **Ranking does**: at whatever rate it fires, fixed-τ escalation beats a random reference at the same realized rate on four of five external pools, paired-bootstrap interval excluding zero (Table 16). The exception is Reasoning QA, where τ fires on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free quantile step: it buys budget control, not discrimination.

Does the gate find failures the expert can fix? The probe is fit and read against local failure, but what a routing system needs is expert benefit: $y=1$ when the local model is wrong and the expert is right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools (Speech TriviaQA 0.796→0.782, Llama Questions 0.830→0.813) and more where the expert is itself weak (Web Questions 0.773→0.689 against a 0.864 expert ceiling; internal 0.840→0.732). The gate thus remains a usable benefit predictor out of distribution (0.63–0.81) despite never having seen an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve. On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but only 65.5% of the failures inside the gate's top-scoring 15% are, and that share climbs monotonically as the cut widens (0.655/0.685/0.711 at 15/30/50%): the gate's most confident failure calls are the ones the expert also misses. A benefit-trained refit closes none of this: with expert outcomes collected for all 2,310 calibration queries (gold-text expert, expert-right rate .853), retraining the same 12,288-d read directly on the benefit label moves internal benefit-AUC .732→.759 (paired +.027 [.004, .052]) and leaves every external pool inside noise (Δ −.029 to +.037, all CIs spanning zero), at no internal cost on the fail label (.840→.839). The benefit gap reflects the label's dependence on the expert's own failure surface, not a trainable objective mismatch: the expert-agnostic label leaves nothing on the table. Table 17 places the deployed gate against the matched-budget oracle that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time or expert-aware second stage would have to claim.

Table 16: One absolute threshold (τ=0.628, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed-τ accuracy minus a random reference at the same realized rate (paired bootstrap over ids, B=10^4, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit (y= local wrong $\wedge$ expert right, base rate in the last column). Rates and accuracies in %; AUC in decimals.

pool	n	rate	local	acc	rand	Δ 95% CI	$\mathrm{AUC}_{\mathrm{fail}}$	$\mathrm{AUC}_{\mathrm{benefit}}$	base
Speech TriviaQA	250	28.4	66.4	83.2	75.0	[+4.4, +12.1]	0.796	0.782	30.8
Speech Web Questions	250	36.0	56.8	72.0	67.5	[+0.8, +8.4]	0.773	0.689	32.4
Llama Questions	250	8.0	83.6	86.8	84.3	[+0.6, +4.6]	0.830	0.813	11.2
SD-QA	200	48.0	51.5	80.5	71.4	[+4.5, +13.6]	0.800	0.765	43.5
Reasoning QA (zh)	202	2.0	58.9	60.9	59.5	[−0.1, +3.5]	0.679	0.634	30.2
internal frozen test	240	35.0	38.3	65.4	57.0	[+3.8, +13.1]	0.840	0.732	53.3

Table 17: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true y=1 cases first. The gate clears random everywhere; the oracle column is the remaining headroom. Accuracies in %.

	15% budget			30% budget			50% budget		
pool	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	71.0	76.0	81.6	75.5	83.6	96.4	81.6	91.2	97.2
Speech Web Questions	61.2	64.0	72.0	65.7	70.0	86.8	71.6	76.4	89.2
Llama Questions	85.0	88.8	94.8	86.4	91.2	94.8	88.2	93.2	94.8
SD-QA	57.7	63.5	66.5	63.9	74.5	81.5	72.2	82.0	95.0
Reasoning QA (zh)	63.1	67.8	73.8	67.4	72.8	89.1	73.0	76.7	89.1
internal frozen test	46.3	49.2	53.3	54.3	60.8	68.3	65.0	74.2	88.3

J Transfer latency shapes

Matched-rate precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail 0.16) the 50% cap is 0.32 and the deployed probe reaches 0.304, 95% of the cap (recall 0.93). Quoting raw precision would misread the strongest pool as the weakest.

The median left-fold. On easy pools per-arm P50 is non-monotonic (Llama Questions 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe preferentially escalates the queries whose local decode is longest: on retrieval pools wrong answers hedge at length (decode time and answer length r=0.97), so a wrongness-selecting probe strips the slow tail as a side effect (escalated-at-15% ids: always-local arm local P50 1.711 s vs. 1.198 s kept; local accuracy 0.32 vs. 0.73). Controls: the probe does not key on length (corr(score, length) $= +0.05$; ≈0 conditioned on wrongness), and on SD-QA, where wrong answers are short, the same probe selects wrongness identically and produces no fold. The mean is monotonic (1.65→2.06 s); kept-local P50 falls 1.52→0.43 s across tiers (Figure 16). On Reasoning QA the shape inverts usefully: the local chain-of-thought is spoken, so every reasoning token enters the response clock, while the expert's chain is hidden behind a flat ∼3.7 s round-trip: escalation truncates the latency tail (P90 13.25→10.94 s) rather than fattening it.

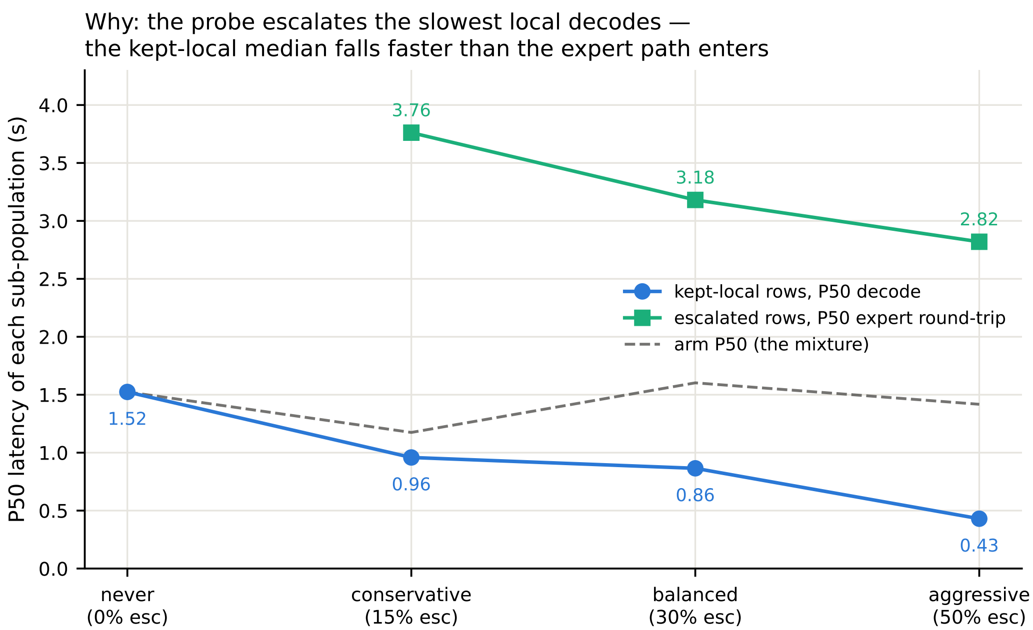


Figure 16: Mixture decomposition of Llama Questions latency per arm: kept-local P50 falls 1.52→0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat ∼3 s expert round-trip; the arm median (dashed) is their mixture.

AlpacaEval scale caveats. The official 4.8 is an offline chat-mode number (our chat-mode control reproduces it at 4.86) and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop's spoken-reply style plus a token cap, so only live arms are comparable to each other. Selected queries score 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71) because the expert writes better long-form answers; random picks would harvest the same gain.

K Full-pool and per-family figures

Figures 17 (dual-view) on the full frozen pool, and the complete external-family figure set (Appendix L); the noise audit behind the ±0.02–0.03 floor is Figure 18.

L External-benchmark figure set

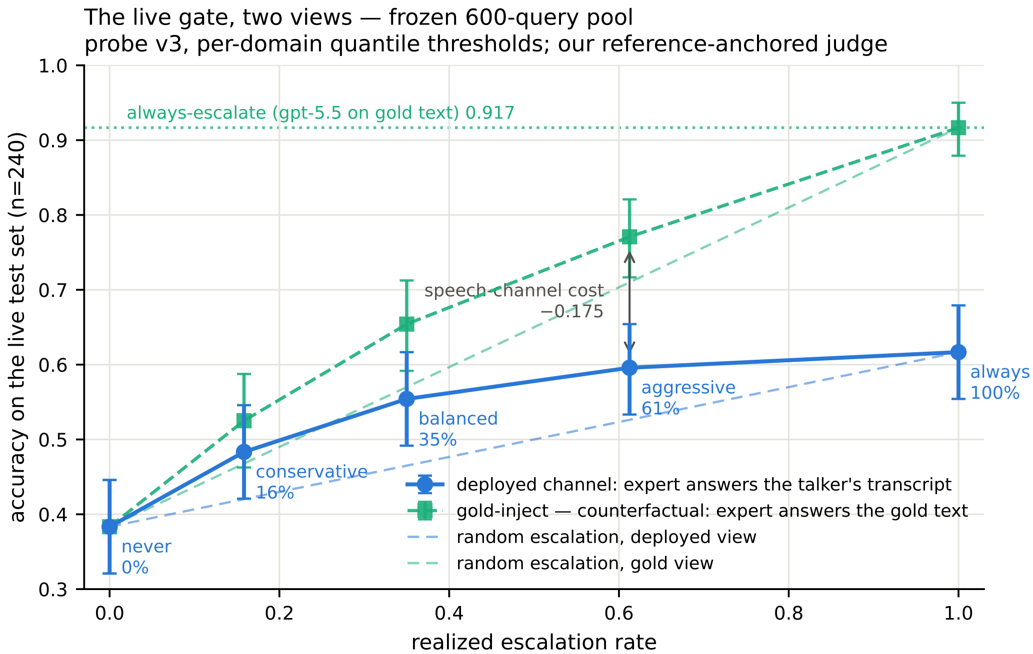


Figure 17: Dual-view live tradeoff, full pool (n=240); companion to Figure 3.

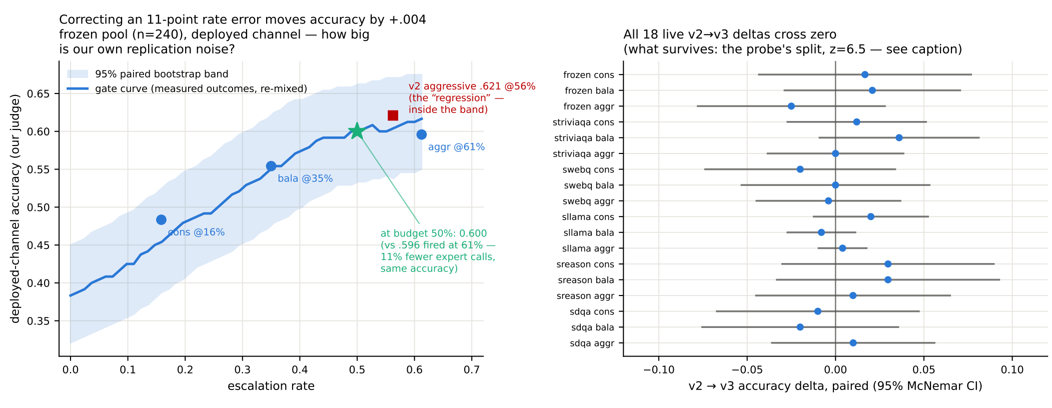


Figure 18: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE 0.009–0.028.

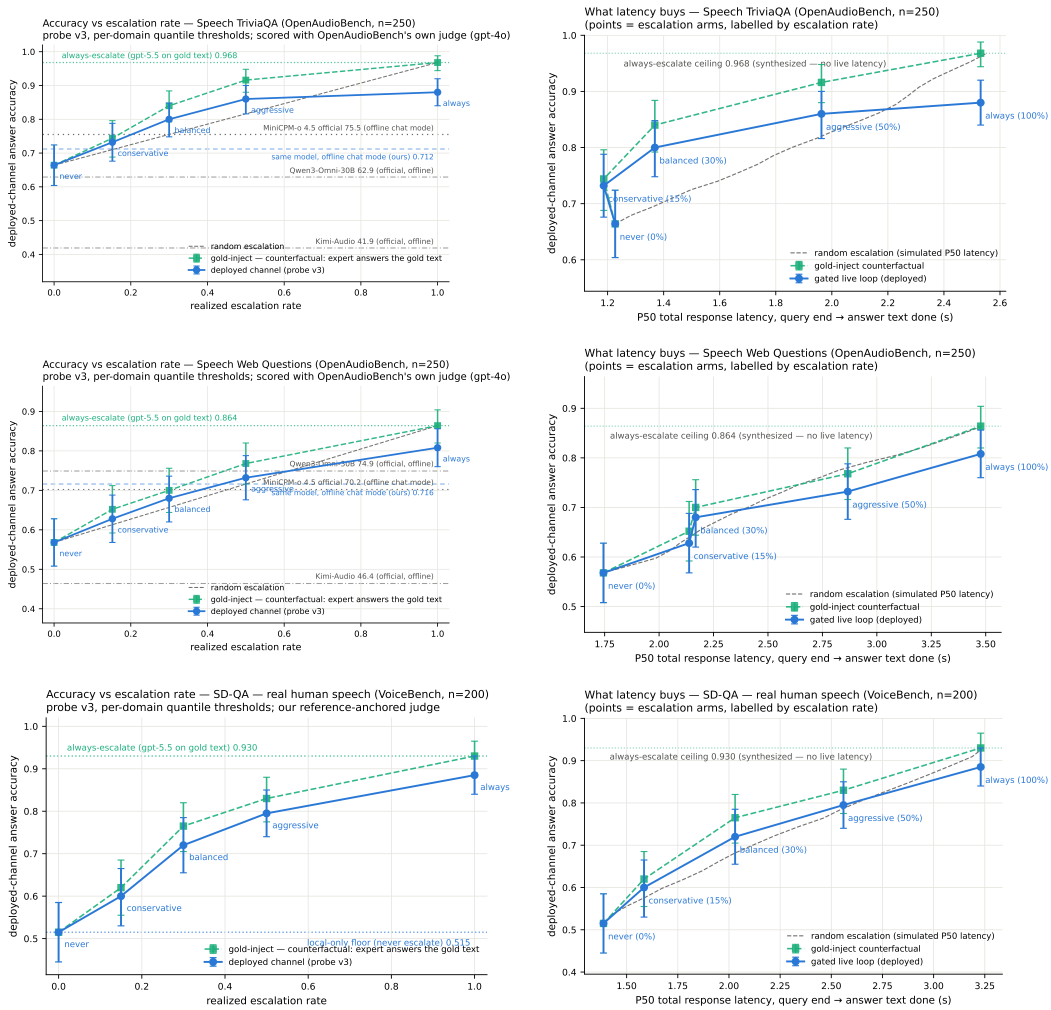


Figure 19: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy–latency (right).

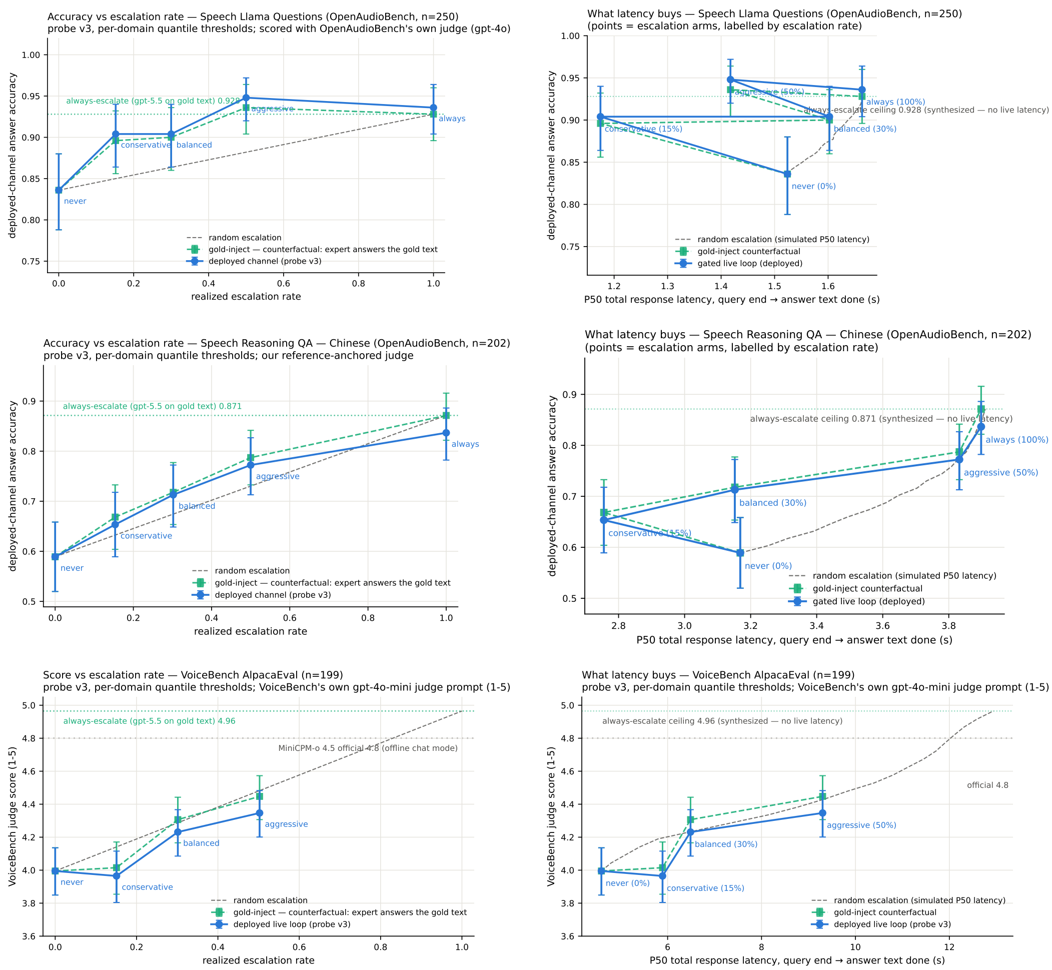


Figure 20: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy–latency (right).

M Judge and self-evaluation prompts

All adequacy labels use the following fixed system prompt (structured outputs with a strict {`reason`, `adequate`} schema; the judge never sees which system produced the answer):

```
You are a strict, fair grader of a small language model's answers.
You are given a QUESTION, an optional REFERENCE answer, and a
CANDIDATE answer produced by a small model.  Decide whether the
CANDIDATE is ADEQUATE, meaning ALL of:  (1) factually correct;
(2) any reasoning shown is valid; (3) it actually and completely
answers the question that was asked.  For multiple-choice questions
the candidate must land on the correct option (by letter or by
unambiguously stating the option's content).  A partially correct,
hedged, refused, off-topic, or truncated answer is NOT adequate.  If
a REFERENCE is provided, grade against it.  Be strict but do not
penalize harmless differences in format or extra correct detail.
```

The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first generated token's distribution:

```
[pre] I am going to show you a question.  Do NOT answer it.  Judge
honestly whether you yourself would answer it correctly.  Question:
{q} Would you answer this question correctly?  Reply with exactly one
word:  Yes or No.
[post] Here is a question and a proposed answer.  Question:  {q}
Proposed answer:  {a} Is the proposed answer correct?  Reply with
exactly one word:  Yes or No.
```

N Reproducibility

All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test split is touched only by the evaluation scripts. Environment pins: torch 2.8.0; transformers 4.51.0 for MiniCPM-o 4.5, with per-model pins for comparison backbones; all experiments run on single-H100 instances. Probe artifacts (weights, feature configuration, per-pool label-free thresholds), the judge prompt above, all trace files, and the per-experiment execution log (including per-run GPU and API spend, on the order of \$800 total) ship in the project repository, released upon publication.

O The native full-duplex regime

Every result so far reads the probe under a turn-based serving loop (harness-controlled end-of-turn) or its harness-interleaved concurrent variant (Appendix H). MiniCPM-o 4.5 also ships a native full-duplex mode: audio prefills in 1 s units into the same context the model generates from, and the head itself emits $\langle$listen$\rangle$ or speaks, yielding the floor via its turn-end token. This is the deployed regime of the demonstration system, and this section recalibrates and re-validates the gate inside it. The read point needs no external end-of-turn detector: the chunk where the head first switches listen$\rightarrow$speak is the commit-to-answer moment, and the probe reads there (the tail-k features then cover the model's first $\sim$8 answer tokens, a generation-side read the head hands us for free). For 87% of test queries this commit lands within $\pm$1 chunk ($\pm$1 s) of the true question end.

Recalibration. The full 2,310-row calibration mix was re-collected in this regime (fresh duplex session per query, features at the deployed read point, labels unchanged) and the same 12,288-d linear read refit. (The final deployed gate goes one step further and also re-labels: each training row's label is the judged outcome of the native answer produced in the same duplex session the features were read from, under the deployed sampling, rather than a separate deterministic turn-based answer. The two label sets agree on 80.8% of the 5,228 rows; the native failure rate is higher, .541 vs. .463, mostly turn-based-correct questions the sampled native answer gets wrong. Ranking is unchanged within noise, Table 18.) On native test-240 features the in-regime probe reaches AUC **0.830** $[0.777, 0.878]$

Table 18: Native-regime probe AUC at the deployed listen→speak read point, under the OFFICIAL serving configuration (final deployed system; the chronological narrative above measured under inherited wrapper defaults). The merged refit adds the 2,700-row public-benchmark expansion (incl. 400 Mandarin rows) to the official core. En-4 mean .770 vs. turn-based .771: the English regime cost is eliminated; the Mandarin slice restores Reasoning-zh from the all-English chance collapse. The last column is the deployed gate: same 5,228 features, but every training label is the judged outcome of the same native-duplex answer the features were read from (instead of a separate deterministic turn-based answer). Label source is a wash for ranking (En-4 mean .771 either way; the paired bootstrap crosses zero on six of seven pools, TriviaQA +.035 [+.006, +.065] being the exception), so the deployed gate uses the in-regime definition. Internal test is scored against both the earlier harness labels and the native-judged labels of the same 240 queries.

pool	n	old cfg	official (2,542)	+exp3 (5,228)	+**native lbl. (deployed)**
internal (harness lbl.)	239	.830	**.846**	.844	.843
internal (native lbl.)	239	—	—	.878	.876
TriviaQA	250	.711	.759	.767	**.802**
WebQ	241	.736	.754	.772	**.777**
Llama-QA	245	.757	.789	**.815**	.807
SD-QA	200	.736	**.737**	.728	.700
Reasoning-zh	202	.606	.495	.605	**.621**

Table 19: Native-regime gated accuracy under the official serving config and the deployed merged (official+exp3) gate (8ad remix: native scores select, native local outcomes + cached always-arm expert outcomes mix). p = permutation vs. matched-rate random; escalation rate in parentheses. Thresholds are the deployed per-language operating points: tier quantiles of the training-set out-of-fold scores, computed separately for the English and the Mandarin training rows (no evaluation pool is consulted); the Mandarin pool uses the zh thresholds. On frozen test the aggressive tier reaches the always-escalate ceiling at 49% of the expert cost. Reasoning-zh, which the global English thresholds never reached (1–2% escalation), now escalates at its nominal rates and beats matched-rate random at the two lower tiers. AlpacaEval (measured under the 2,542-row official gate) stays a clean honest negative.

pool	local	balanced	aggressive	always
frozen test	.431	.565 (27%, p=.0005)	**.669** (49%, p<.0001)	.669
TriviaQA	.568	.684 (20%, p=.0005)	.852 (54%, p<.0001)	.916
WebQ	.411	.527 (22%, p=.045)	.680 (58%, p=.020)	.809
Llama-QA	.767	.792 (4%, p=.002)	.841 (16%, p<.0001)	.910
SD-QA	.455	.605 (24%, p=.0035)	.775 (61%, p=.003)	.895
Reasoning-zh (zh thr.)	.525	.658 (34%, p=.023)	.713 (55%, p=.066)	.807
AlpacaEval (VB 1–5)	3.65	3.70 (6%, n.s.)	4.07 (37%, p=.24)	4.76

(transferred concurrent probes: 0.807–0.818; turn-based reference 0.877). Externally the in-regime refit beats both transferred probes on all five pools: TriviaQA .711 (Δ+.080 [+.030,+.129]), WebQ .736, Llama-QA .757 (Δ+.118), SD-QA .736, Reasoning-zh .606 (Δ+.115); external mean .709. The regime ordering is monotone everywhere: turn-based (.877/.771 internal/external mean) > native (.830/.709) > concurrent (.689 external), i.e. a substantial share of the concurrent regime's cost was the harness read point, not liveness itself. Thresholds again do not transfer (balanced quantile 0.386→0.645); scores are decoding-temperature invariant (0.7-trained probe on temp-0.1 features: AUC 0.843).

Validity. Re-mixing native local outcomes (gpt-judged, deployed decoding) with cached always-arm expert outcomes under the native scores (Table 19): under the deployed gate, gated accuracy clears matched-rate random on all five English pools at both deployed tiers: frozen .431→.669 at 49% escalation (p<.0001; ceiling .669), TriviaQA .568→.852 at 54%, WebQ .411→.680, Llama-QA .767→.841, SD-QA .455→.775 at 61%. In-regime training renormalizes the score distribution, so realized escalation rates sit near their nominal quantiles on the English pools (the earlier 64–73% drift was a transferred-threshold artifact). Reasoning-zh is an operating-point problem before it is a ranking problem: a single global threshold, quantiled on a 92%-English training mix, places the zh score distribution below every tier (1–2% fire at "balanced"), the language axis of Appendix H expressed as a fire-rate failure. The deployed fix needs no evaluation data: tier thresholds are quantiled on the training-set out-of-fold scores per language (en/zh), and the session selects the language's thresholds. The zh thresholds derived from the 400 Mandarin training rows alone land Reasoning-zh at 17/34/55% fire (nominal 15/30/50) and lift it from .525 to .609/.658/.713, beating matched-rate random at the two lower tiers (p=.0065, p=.023). This static per-language point is only as good as the match between the language slice's family mix and the deployment stream: when the Mandarin training slice is widened from 400 to 1,900 rows with higher-failure families (below), its balanced quantile moves from .39 to .75 and fires 5% on Reasoning-zh; a family-balanced quantile recovers 13/24/38%. The composition-free alternative (each pool's own score quantile, or its online form, a 100-turn sliding-window tracker with no labels and no pool identity) realizes nominal rates on every pool and converges within $\pm$.06 of nominal on the English pools.

Is the Mandarin gap a language gap? No: it is a task-type gap. A targeted 1,500-row Mandarin expansion (Chinese SimpleQA long-tail facts, C-Eval and CMMLU knowledge MC, the remaining XCOPA-zh; public benchmarks, deduplicated against every pool, judged on the native answers; native failure .578) does not move Reasoning-zh when added to the calibration mix (.621$\rightarrow$.627, paired +.006 [$-$.026, +.040]; internal test $-$.010, English external +.006), and Mandarin-only training is worse everywhere (Reasoning-zh .591, internal .758). Read the other way, the deployed gate scores these 1,500 unseen Mandarin questions at AUC **.811** [.789, .833] (long-tail .719, C-Eval .730, CMMLU .642, causal .693), the same band as the English pools. What the probe does not read is multi-step reasoning, which is what Reasoning-QA is; the ".62 on Mandarin" number is that gap wearing a language label. The expansion is therefore not deployed and serves as a second, larger Mandarin held-out pool. (A side observation from the same dump: on the causal-commonsense family 34% of the native answers switch to English while remaining judged adequate, a language-consistency defect the adequacy label does not see.)

Reasoning coverage and in-context rows: the read point saturates. Two further targeted sets test the two remaining hypotheses. A bilingual multi-step reasoning pool (1,319 rows: LogiQA and LogiQA-zh, MATH levels 4–5 without LaTeX, StrategyQA, BBH date/deduction/causal/temporal/navigation, CMATH grades 4–6, Ape210K; native failure .595) leaves Reasoning-zh unchanged when added to calibration (.621$\rightarrow$.620, paired $-$.002 [$-$.039, +.038]) and moves nothing else beyond noise. The reason is visible inside the pool itself: cross-validated AUC on the reasoning families is only .61–.76 (overall .736 [.708, .762]), against .77–.82 for fact, knowledge-MC, and causal families. Whether a multi-step chain will come out right is weakly encoded in the listen$\rightarrow$speak commit state, which precedes the chain; the .62 on Reasoning-zh is that ceiling, not a coverage gap. Second, the calibration questions re-collected as the second turn of a session (a spoken carrier question and its answer first) fail more often (.517$\rightarrow$.620; 26% of labels flip) while the deployed gate scores them lower (mean shift $-$.097): at the balanced tier the in-context escalation rate halves (.252$\rightarrow$.115) although ranking holds (AUC .861 [.815, .896]). Adding these rows to training does not change AUC either; the defect is the operating point, and the deployed demo therefore thresholds on a sliding-window quantile of the session's own recent onset scores (100-score window, static per-language threshold until 20 scores exist), which follows such shifts without labels or pool identity. After a label-definition fix and three targeted expansions ($\approx$4,000 rows), external ranking sits where the 5,228-row gate left it. The head is not the bottleneck either: on the same features, per-dimension standardization costs .07 of English external AUC (the raw activation scale is informative), an elastic-net head is a wash (internal +.008, external $-$.007), and non-linear heads are clearly worse (256-unit MLP .702, PCA-256 + gradient-boosted trees .715, vs. .771). Nor are several heads: a three-way task-type router trained on the same features is 97% accurate, yet no family-specialized head beats the single head on its own family (facts .846 vs. .847, reasoning .761 vs. .766, chat .672 vs. .720), and the routed mixture is flat-to-negative externally (SD-QA $-$.033 [$-$.060, $-$.007]). What the commit-chunk state encodes about the coming answer is extracted by one linear read; the remaining headroom lies in what is read: later into the answer, other layers, or a second signal family such as verbalized self-evaluation (Table 7), not in how. The talker's own local floor is lower in-regime (frozen .483→.431 under the official serving configuration; the inherited-defaults harness measured .371, and the ≈one-third-temperature / ≈8-point decomposition belongs to that older figure), which raises the gate's marginal value in deployment. The in-regime calibration scaling curve (measured under the default serving configuration) mirrors the concurrent one and is not saturated: internal .717→.830, external mean .643→.709 from 360→2,310, and a further public-benchmark expansion to 5,009 rows (2,300 English across the same seven families + 400 Mandarin: MGSM-zh and XCOPA-zh through the same TTS pipeline, source-disjoint from Reasoning-zh) continues the $\approx +.02$-per-doubling external trend to **.736** while internal stays flat (.834), and the Mandarin slice moves the axis English data never touched, Reasoning-zh .606→.659 (Δ vs. conc-2310 $+.169$ [$+.085$,$+.251$]; the deployed official-configuration merge reads the same pool at .605 with turn-based labels and .621 with the deployed native labels, Table 18). Two caveats on reading this curve: its six points are the project's chronological accumulation, each step adding new query families, coverage, or a serving-configuration change, not repeated random subsamples of the final set, so it evidences that added coverage transfers, not that another random doubling of the same families would return $+.02$; and the internal/English-external plateau against the turn-based reference (.771) is an empirical baseline at n≈200–250 per pool, not a ceiling.

Ceiling, stability, and the pre-answer read. Three diagnostics on the deployed 5,228-row gate. (i) **Ceiling.** A pool-prior scorer (each query scored by its training pool's out-of-fold failure rate) reaches AUC .786 against the probe's .848 on the training out-of-fold scores, and .767 against .876 on frozen test; AUC restricted to same-pool pairs is .743 (train) and .820 (test). Per-question discrimination is thus $+.06$–.11 of AUC on top of the type prior, the same decomposition as the n=600 audit of Section 3.1; at a 30% budget the prior alone recovers .475 failure recall against the probe's .506 (random .299). The external pools, each a single source, measure the within-pool component directly (.70–.81, Table 18). (ii) **Stability.** An independent second native pass over frozen test (deployed sampling, T=0.7, its own judged label) changes 98% of answer texts and flips 14% of native labels, yet the read at the commit chunk barely moves: feature cosine .95, score correlation .95, and the tier decisions agree on 91–94% of queries. Each run's features predict the other run's outcome about as well as their own (.848/.887 vs. .876/.872), so the probe reads a property of the question in context rather than the fate of one sampled answer; against the 204 queries whose two outcomes agree, AUC is .91–.93, which bounds the single-sample label noise in the reported numbers. (iii) **Pre-answer read.** The same layer-22 read taken before the onset chunk's generate call (no answer tokens in the tail), scored by the deployed head without refitting, gives AUC .861/.837 against the two label sets, .011–.015 below the deployed read; the first ≈8 answer tokens are worth about one AUC point, and the pre-answer variant needs its own operating points (the deployed quantiles fire 0/1/20% on it).

Live validation. Full live runs on frozen test-240 (real raw-audio ASR uplink, real expert, wait paced at one chunk per second) validate the remix arithmetic where the channel is thin and expose where it is not: balanced delivers .504 at 23% fire (remix predicted .492 at 24%); aggressive delivers .537 at 45% against a .596 remix prediction, and the gap decomposes entirely onto fired turns: live fired-turn accuracy .495–.589 vs. the .667 cached-expert ceiling. The native relay channel (the answer re-entering as a text unit and being voiced) is the regime's lossy element (89% of relay units need the nudge retry), playing the role the transcription uplink played in the turn-based loop; the uplink itself is now clean (raw-audio ASR, p50 0.9 s). Latency, expert-cache warm: expert p50 4.4–5.7 s, wait p50 5–6 chunks, fire→relay first audio p50 ∼7–8 s. Selection is visible live: unfired subsets score .478/.573 against the .371 unconditional local floor.

Dialogue-act gating. Full duplex creates commits that are not answers: after a barge-in or a backchannel the head speaks to manage the floor ("Okay."), and the failure probe is out-of-distribution there, on 194 TTS'd floor-management stims (stop commands, backchannels, acknowledgments, fillers; en+zh) the unmodified gate false-fires on 37% at the aggressive tier (stop commands 45%), i.e. it would escalate "stop" to the expert. The fix is a second linear head on the same L22 read, info-seeking vs. floor-management, which separates the two acts perfectly (OOF AUC 1.000); thresholded at the question-side 0.5th percentile it costs 0.5% of true escalations and admits 0% of floor stims. Escalation requires both heads: $\mathrm{act} \geq \tau_a \wedge P(\mathrm{fail}) \geq \tau$. The deployed demo answers floor turns naturally and reports the act score per commit; the failure probe's calibration is untouched.

Live use then sharpened both heads within minutes: scripted pools miss what ten minutes of real conversation finds. (a) Live mic speech shifts both act distributions into the calibration gap (real request-phrased questions score ∼.93, live floor turns ∼.37 vs. a standalone-stim maximum of .05): a percentile-anchored τ_a rejected a real question and the talker delivered an empty promise. The fix is a gap-center threshold plus in-context calibration stims, each utterance dumped as the second turn of a carrier conversation, matching how floor acts actually arrive; the act read stays perfectly separable (AUC 1.000) under both conditions, and in-context stop commands false-fire the failure probe on 57% at the aggressive tier, so the act term is not optional. (b) The native refit had silently dropped the FreshQA real-time extension ("the stock price of NVDA today" scored $P(\mathrm{fail}){=}.29$ and stayed local); replaying that recipe in-regime restores fast-changing heldout fire to .45/.75/1.00 across tiers at zero internal-AUC cost (.830→.833), with budgets still quantiled on the core mix. (c) A relay's own speak-onset must not re-enter the gate (re-fire with an empty uplink); deliveries are guarded to their end of turn. The pattern across all three: the mid-layer signals separate cleanly every time, and what re-opens at each regime or domain shift is calibration coverage and thresholds: interactive use is itself an evaluation modality.

Multi-turn grounding. A stateless expert uplink breaks on follow-ups: "what is NVDA trading at" then "what about Apple" sends only the second utterance, unresolvable in isolation. The probe itself is unaffected: it reads L22 with the prior turns live in the model's KV cache, so the fix is in the uplink, not the gate: the deployed demo threads a rolling resolved-dialogue history into the expert input and asks it to resolve references, after which the follow-up correctly returns Apple's share price. The measured eval pools are single-turn, so this is a deployment capability rather than a table; systematic multi-turn evaluation is future work.

Floor control, natively. With the harness gone, 8ba's question re-poses on the head itself (108 scripted overlap trials, stims spoken over the talker at fixed offsets). During answer speech, backchannels almost never stop the floor (false-stop 3/24 within a 6-chunk window) with no ASR or lexicon anywhere in the loop; sustained question speech yields the floor (6/12 within 6 chunks, median 8.5 post-stim); short burst commands ("Stop!") mostly do not (3/12): the native head reads sustained speech, not commands, a naturalness-for-reliability trade the old energy-VAD did not make. The escalation machinery is floor-transparent where it should be: relays complete under overlapping stims (26/30), and backchannels during the thinker wait leave the pending relay intact (8/10). The weak stop-command sensitivity decomposed into two causes, found in sequence. **Mechanism**: instrumenting the serving loop shows the head never samples $\langle$listen$\rangle$ mid-turn (zero attempts across 36 trials), so its only stop channel is the turn-end token. **Configuration**: a user-demanded vanilla control (the bare official serving stack, no gate machinery) revealed that our loop had inherited the checkpoint wrapper's sampling defaults ($k{=}100$) rather than the official demo's serving config ($k{=}20$, a 3-chunk startup listen guard, an assistant-style system prompt). Under the official config the turn-end channel is responsive: a spoken "stop" two seconds into an enumeration answer ends the turn in 2.0–2.2 s in 3/3 scripted sessions, versus 1–13 s (once running the answer to completion) under the inherited defaults: wide-k sampling dilutes away the rising turn-end probability. The deployed demo now serves the official config, and the earlier floor-control stop numbers measured under the inherited defaults should be read as a lower bound on native responsiveness. Two durable lessons: serving-parameter parity with the reference stack is part of "native", and a vanilla control arm belongs in the harness from day one. The one true interaction: a new question during the wait derails the pending relay 7/10 times: the head simply moves on, which is the empirical case for aborting the in-flight expert call on user speech, left as deployment future work. Latency: listen chunks cost 0.04 s and speak chunks ∼0.5 s on one H100 (realtime holds); fire→stall-audio 0.2–1 s; fire→relay-first-audio is the expert wall clock (13–20 s observed) plus ∼2.5 s.